

Clinical Investigation Plan (CIP)

Title: EpiSave ambulatory: A multi-center, event-driven clinical investigation to assess the real-world performance of the EpiSave generalized convulsive seizure detection system and the caregiver alert system in people with epilepsy.

Short title: EpiSave ambulatory: Outpatient validation study

Type of investigation:	Clinical investigation concerning medical devices (interventional)
Categorisation:	Category C2
Registration:	To be determined
Identifier:	To be determined
Sponsor-Investigator:	Centre Hospitalier Universitaire Vaudois (CHUV) Address: Rue du Bugnon 46, 1011 Lausanne, Switzerland Contact Person: Prof. Dr. med. Philippe Ryvlin Email: philippe.ryvlin@chuv.ch Phone: +41 21 314 39 66
Sponsor representative (if the Sponsor is not located in Switzerland)	NA
Medical Device (MD):	Name of the MD: EpiSave system EpiSave system is composed of the EpiSave algorithm, a proprietary Generalized Convulsive Seizures (GCS) detection algorithm developed by the research group NeuroDigital@NeuroTech embedded in the EpiSave smartwatch app running on Android Wear OS smartwatches with a paired EpiSave smartphone app.
CIP Version and Date:	CIP Version: 1.0 CIP Version Date: 16.12.2025

Signature Pages

ID number of the investigation: Investigation ID
Registry and registration number

Title: **EpiSave ambulatory:** A multi-center, event-driven clinical investigation to assess the real-world performance of the EpiSave generalized convulsive seizure detection system and caregiver alert system in people with epilepsy.

The Sponsor, the Principal Investigator and the Statistician have approved the CIP version 1.0 (dated 16. 12.2025) and confirm hereby to conduct the investigation according to the CIP, the current version of the World Medical Association Declaration of Helsinki, ISO14155:2020 norm, ICH-GCP as far as applicable, and the local legally applicable requirements.

The Investigator has received the ICF and consider it appropriate for use.

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Place/Date

Signature

Statistician: Mireille Moser

Place/Date

Signature

Principal Investigator at the local investigational site:

I have read and understood this CIP version v1.0 (dated 16.12.2025), and agree to conduct the investigation according to the CIP, the current version of the World Medical Association Declaration of Helsinki, ISO14155:2020 norm, ICH-GCP as far as applicable, and the local legally applicable requirements.

I have received the ICF and consider it appropriate for use.

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SYNOPSIS

Sponsor / Sponsor-Investigator	Centre Hospitalier Universitaire Vaudois (CHUV)
Title:	A multi-center, event-driven clinical investigation to assess the real-world performance of the EpiSave generalized convulsive seizure detection system and caregiver alert system in people with epilepsy.
Short title / Investigation ID:	Short title: EpiSave ambulatory: Outpatient validation study Investigation ID: to be assigned
Clinical Investigation Plan, version and date:	CIP Version: 1.0 Date: 16/12/2025
Registration:	ClinicalTrials.gov: [Registration number to be assigned] EU-CTR: [Registration number to be assigned] Swiss National Portal: HumRes
Category and its rationale:	Category: Medical Device Category C (Sub-category C2) Risk Level: Minimum risk Justification: This clinical investigation involves the EpiSave system, which is not CE-marked as a medical device under Article 13 MedDO. Although the hardware platform (Wear OS Android smartwatch such as TicWatch Pro 5) is CE-marked, the investigational software used for seizure detection is not certified for medical use. Therefore, the study falls under Category C, specifically Sub-category C2, as per Art. 6(3)(b) ClinO-MD: “The device to be investigated does not carry a conformity marking in accordance with Article 13 MedDO.” The investigational use of the EpiSave system is interventional and aims to assess its performance and safety in real-world ambulatory settings. Approval from Swissmedic is required under the ClinO-MD ordinance.
Name of the MD, Unique Device Identification (UDI), name of the manufacturer	Name of the MD: EpiSave system EpiSave system englobes the EpiSave smartwatch App with proprietary GCS detection algorithm (EpiSave algorithm) developed by the NeuroDigital@NeuroTech research group and the EpiSave smartphone companion App. The Apps have been developed by the Italian company UQIDO.
Stage of development:	Pivotal stage The purpose of this study is to generate robust clinical data on the performance of the EpiSave system and its safety.

Background and rationale:	Epilepsy affects approximately 60 million people worldwide, with 25-30% experiencing GCS. GCS represent one of the most severe forms of epileptic seizures and are associated with increased risks of physical injury, sudden unexpected death in epilepsy (SUDEP), and significant impact on quality of life. Current medically certified seizure detection solutions suffer from critical limitations: - High false-alarm rates (>1/week) - Substantial costs (\$250-1,500 USD plus monthly fees) - Limited availability (night-only operation for some devices) - Stigmatizing dedicated wrist devices - Lack of reimbursement in most countries The EpiSave system addresses these limitations through an innovative Android-based smartwatch app and its paired smartphone app designed to detect GCS with high sensitivity and low false alarm rates (FAR). The development follows a standardized framework, with EpiSave algorithm having successfully completed phase 2 testing, which involved algorithm development and offline validation on data previously collected in EMUs where video-EEG serves as a gold standard.
Objectives:	Primary objective: To assess EpiSave algorithm sensitivity for video-confirmed nocturnal GCS detection in real-world ambulatory settings Secondary objective: evaluate the system's performance in detecting GCS during different time periods, evaluate FAR, caregiver response, GCS and its phases duration precision, device usability, quality of life impact
Outcomes:	Primary outcome: EpiSave algorithm sensitivity for video-confirmed nocturnal GCS detection Secondary outcomes: FAR, caregiver response time, GCS and its phases duration precision, device usability score, quality of life Impact
Design:	Prospective, interventional, multi-centre, event-driven clinical investigation, single-arm, open-label study
Inclusion / exclusion criteria:	Patients eligible for inclusion in the study must meet all of the following criteria: be ≥ 6 years old at the time of enrolment; have a confirmed diagnosis of drug-resistant epilepsy documented by a neurologist; ≥2 nocturnal GCS per year; be able to wear and feel comfortable wearing a smartwatch; own a compatible smartphone (Android); have a caregiver living nearby who can intervene quickly if needed (within xx minutes of the onset GCS); provide informed consent (or have it provided by a legal guardian, if applicable). Participants will be excluded if they meet any of the following criteria: Age < 6 years old; uncertain or unconfirmed diagnosis of drug-resistant epilepsy; <2 nocturnal GCS per year; not be able to wear or not feel comfortable wearing a smartwatch; do not own a compatible smartphone (Android); haven't a caregiver living nearby who can intervene rapidly if needed (within xx minutes of the onset GCS); unable to provide informed consent or no legal representative available. Vulnerable Participants: The EpiSave ambulatory study includes paediatric participants (6-17 years) and those with cognitive limitations, as these vulnerable populations are particularly at risk of SUDEP, especially due to their inability to alert others or reposition themselves after a GCS. Their inclusion is essential to evaluate the effectiveness of EpiSave GCS detection system, as their specific needs cannot be studied indirectly.

Measurements and procedures:	The participant must wear on the wrist of the non-dominant arm the smartwatch day and night, with the EpiSave system active. The EpiSave smartwatch app detects GCS in real time (thanks to the EpiSave algorithm) and automatically sends an SMS alert to the designated caregiver via the EpiSave smartphone app, notifying them of the GCS event and the patient's geographic location. To confirm episodes of nocturnal GCS in the home environment, participants receive a dedicated night-time recording camera. Daytime seizures can, in most cases, be confirmed by relatives or witnesses. Patients will have monthly follow-up phone calls to check the functioning of the EpiSave system, the smartwatch, and the surveillance camera. These calls will last about 10 minutes. The patient completes questionnaires on quality of life in epileptic patient (QOLIE-31) and device usability at the end of the study.
Intervention:	Duration of intervention: 6 months per participant, up to 24 months total recruitment period. Mode of Action of the EpiSave algorithm: - Continuous monitoring of motion data via smartwatch 3D-accelerometry (3D-acc) sensor - Real-time analysis using proprietary GCS detection algorithm - Upon GCS detection: collection of additional biosignals for 15 minutes post-GCS. Automated dual-alert system to patient and caregiver (by sending a text message)
Control intervention (if applicable):	No Active Comparator.
Number of subjects with rationale:	Recruitment of up to 100 patients
Duration of the investigation:	30 months
Investigation schedule:	First- subject –In (planned): Month 3 Last- subject –Out (planned): Month 18-20
Investigator(s):	<u>Principal investigator:</u> Prof. Dr. med. Philippe Ryvlin Centre Hospitalier Universitaire Vaudois (CHUV) Address: Rue du Bugnon 46, 1011 Lausanne, Switzerland Email: philippe.ryvlin@chuv.ch Phone: +41 21 314 39 66 <u>Co-investigators:</u> Pr. G Ramantani; Pr. S. Vuillemoz; Dr. M. Hardmeier; Pr. A. Datta; Dr. C. Friedrichs-Maeder; Dr. M. Galovic; Dr. L. Imbach; Dr C. Prosperetti; Dr. O. Henning; Pr. C. Baumgartner; Pr. E. Trinkä; Pr. S. Beniczky; Pr. S. Rheims; Pr. M. Holtkamp; Pr. C. Meisel; Pr. R. Surges; Pr. F. Rosenow
Investigational Sites:	Multicentric multinational investigation with: - 9 centres in Switzerland (Lausanne: 1, Geneva: 1, Basel: 2, Berne: 1, Zürich: 3, Lugano: 1) - 4 centres in Germany (Frankfurt: 1, Bonn: 1, Berlin: 2) - 2 centres in Austria (Vienna, Salzburg) - 2 centres in Denmark (Copenhagen, Dianalund) - 1 centre in France (Lyon) - 1 centre in Norway (Oslo)

Statistical considerations:	Based on published recommendations for phase 4 seizure detection studies Power Calculation: Provides 90% power to detect sensitivity $\geq 90\%$ with 95% confidence Event-Driven Design: Ensures adequate statistical power regardless of recruitment timeline Dropout Consideration: Accounts for 20% anticipated dropout rate Recruitment Timeline: 24 months (≤ 1 PWE/4 months/centre)
Compliance statement:	This investigation will be conducted in full compliance with the CIP, the current version of the Declaration of Helsinki, ISO 14155:2020 ICH-GCP (as far as applicable) as well as all national legal and regulatory requirements.

ABBREVIATIONS

AE	Adverse Event
ADE	Adverse Device Effect
ASADE	Anticipated Serious Adverse Device Effect
ASMs	Anti-Seizure Medications
CA	Competent Authority (e.g. Swissmedic)
CE	European Conformity marking
CEC	Competent Ethics Committee
CIP	Clinical investigation plan
ClinO	Ordinance on Clinical Trials in Human Research (<i>in German KlinV, in French Oclin, in Italian OSRUm</i>)
ClinO-MD	Ordinance on Clinical Trials with Medical Devices (<i>in German: KlinV-Mep, in French: Oclin-Dim, in Italian: OSRUm-Dmed</i>)
CNN	Convolutional Neural Network
CRF	Case Report Form (pCRF paper CRF; eCRF electronic CRF)
CS	Convulsive Seizure
DD	Device Deficiency
DEE	Developmental and Epileptic Encephalopathies
DMC / DSMC	Data Monitoring Committee, Data Safety Monitoring Committee
EEG	Electroencephalogram
EMU	Monitoring Epilepsy Unit
FADP	Federal Act Data Protection
FAR	False Alarm Rate
FBTCS	Focal to Bilateral Tonic-Clonic Seizures
FDA	Food and Drug Administration
GCS	Generalized Convulsive Seizures
GDPR	EU General Data Protection Regulation
GTCS	Generalized Tonic-Clonic Seizures
Ho	Null hypothesis
H1	Alternative hypothesis
HRA	Human Research Act
HumRes	Swiss Human Research Portal
IB	Investigator's Brochure
ICF	Informed Consent Form
ICH-GCP	International Council for Harmonisation – guidelines of Good Clinical Practice
IECs	Independent Ethics Committees
IFCN	International Federation of Clinical Neurophysiology
IFU	Instruction For Use
ILAE	International League Against Epilepsy
IRBs	Institutional Review Boards
ISF	Investigator Site File
ISO	International Organisation for Standardisation

ITT	Intention to treat
MedDO	Medical Devices Ordinance (<i>in German: MepV, in French: Odim, in Italian: Odmed</i>)
MD	Medical Device
MDR	Medical Device Regulation (EU) 2017/745 of 5 April 2017
PGES	Postictal Generalized EEG Suppression
PI	Principal Investigator
PP	Patient partner
PPG	Photoplethysmography
PWE	People With Epilepsy
QOLIE-31	Quality Of Life in Epilepsy Inventory
SADE	Serious Adverse Device Effect
SAE	Serious Adverse Event
SDV	Source Data Verification
SNSF	Swiss National Science Foundation
SUDEP	Sudden Unexpected Death in Epilepsy
SOP	Standard Operating Procedure
TWP5	TicWatch Pro 5
USADE	Unanticipated Serious Adverse Device Effect
v-EEG	video-electroencephalography

SUMMARY OF THE REVISION HISTORY IN CASE OF AMENDMENTS

Version Nr, Version Date	Chapter	Description of change	Reason for the change

Investigation schedule

Investigation Periods	Screening	Follow-up visits each month by phone during monitoring with smartwatch and camera					End of study
Visit number	1	2	3	4	5	6	7
Time (week)	0	4	8	12	16	20	24
Visite duration (minutes)	60	5-10	5-10	5-10	5-10	5-10	60
Written consent (ICF)	x						
In- /Exclusion Criteria	x						
Medical History	x						
Demographics	x						
Smartwatch and surveillance camera provided to participant	x						
Participant and caregiver training on the use of the study equipment	x						
System validation and testing	x						
Confirmation of alert system functionality	x						
QOLIE-31 questionnaire	x						x
Fatigue/Seizure Recovery Questionnaire		After the occurrence of a GCS					
Compliance assessment		x	x	x	x	x	x
Adverse Events, Adverse device effects		x	x	x	x	x	x
Device performance evaluation		x	x	x	x	x	x
Data collection verification		x	x	x	x	x	x
Device Deficiencies		x	x	x	x	x	x
Smartwatch and surveillance camera return							x
Download and deletion of data from the smartwatch							x
SUS questionnaire							x
Custom EpiSave system questionnaire							x

1. INVESTIGATION ADMINISTRATIVE STRUCTURE

1.1 Sponsor, Sponsor-Investigator

The sponsor of this investigation is the CHUV | Centre hospitalier universitaire vaudois, represented by the clinical trial unit (CTU). The CHUV is responsible for initiating, designing, and overseeing the clinical investigation, ensuring regulatory, ethical, financial, and legal compliance. They manage data collection, monitoring, and analysis, and ensure the integrity of results. The sponsor also prepares and submits the final clinical report. If acting as Sponsor-Investigator, they additionally conduct the study directly at the investigation site.

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Mireille Moser, holding a Master's degree in Mathematics, is affiliated with the Centre Hospitalier Universitaire Vaudois (CHUV) and contributes to the study through statistical analysis and consultation on study design.

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1.4 Laboratory

Not applicable.

1.5 Monitoring institution

The investigation will be monitored by the Clinical Research Unit – Centre de Recherche Clinique of the CHUV.

1.6 Data Safety Monitoring Committee

Not applicable.

1.7 Any other relevant Committee, Person, Organisation, Institution

Not applicable.

2. ETHICAL AND REGULATORY ASPECTS

The final positive decision of the CEC and of the CA on the conduct of the investigation will be made and given in writing to the Sponsor before the investigation can start (Category C investigations: In Switzerland the national approval is issued by Swissmedic and includes the approval of the CEC). Additional requirements set by the authorities must be implemented.

2.1 Registration of the investigation

Registry	Registration number	Date
ClinicalTrials.gov	NCT	Not obtained yet
EU Clinical Trials Register (EU-CTR)	EudraCT	Not obtained yet
Swiss National Portal	HumRes	Not obtained yet

2.2 Categorisation of the investigation

The investigational device used in this study is the EpiSave system that comprised the EpiSave algorithm embedded in the EpiSave smartwatch app, designed to detect GCS and alert caregivers in real time via the paired EpiSave smartphone companion app. While the smartwatch hardware platform (TicWatch Pro 5, Mobvoi) and the companion smartphone are CE-marked, the EpiSave system itself is an investigational system that does not bear a conformity mark as a medical device.

Furthermore, the system is used in a clinical investigation setting and is not yet approved for medical use in Switzerland. Therefore, the study falls under Category C, specifically Subcategory C2, as defined in Art. 6(3)(b) ClinO-MD: "The product does not bear a conformity mark according to Article 13 MedDO." According to Medical Device Regulation, EpiSave is considered a class IIa non-invasive medical active device/software that monitors physiological processes (rule 11 software) without vital risk.

This categorisation requires submission to Swissmedic and approval prior to study initiation. The classification reflects the investigational nature of the software and ensures appropriate regulatory oversight for safety and performance evaluation.

2.3 Competent Ethics Committee (CEC)

The Sponsor-Investigator will submit the investigation to the CEC and obtain ethical committee approval before the start of the investigation. Each PI at each participating investigational site ensures that approval from their local CEC is obtained and filed in the Investigator site file before the investigation starts.

Reporting duties to the Competent Ethics Committee

Amendments are reported according to Art. 15 ClinO-MD (see also 2.10).

The regular or premature end of the investigation as well as the interruption of the investigation is reported to the CEC within 15 days (within 24 hours if it is due to security reasons) (Art. 36 ClinO-MD). The reasons for a premature end or an interruption have to be explained.

A final report shall be submitted within one year after the regular end of the investigation and within 3 months after a premature end of the investigation. A summary in easily understandable terms must be included with the final report (Art. 37 ClinO-MD).

2.4 Competent Authorities (CA)

The Sponsor-Investigator will submit the investigation to the CA and obtain regulatory approval before the start of the investigation. Each PI at each participating investigational site ensures that approval from the CA is obtained and filed in the Investigator site file before the investigation starts.

Reporting duties to the competent authorities

The reporting duties and allowed time frame for the reporting to the CA include the reporting duties in case of planned or premature end of the investigation and the final report. Reporting duties and timelines are the same as for CEC, except of non-substantial amendments that shall be reported as soon as possible (Art. 20 ClinO-MD). For safety reporting including reporting of safety and protective measures, as well as the safety and general progress report (jointly "annual report") refer to chapter 10. Amendments to the CIP are reported according to Art. 20 ClinO-MD (see also 2.10). A final report shall be submitted within one year after the regular end of the investigation and within 3 months after a premature end of the investigation. A summary in easily understandable terms must be included with the final report (Art. 38, resp. 37 ClinO-MD).

The regular or premature termination of the investigation as well as the interruption of the investigation is reported to the CA within 15 days and within 24 hours if it is due to safety reasons (Art. 36 ClinO-MD). The reasons for a premature termination or an interruption shall be explained.

A final report is submitted within one year after the regular end of the investigation and within 3 months after a premature end of the investigation. A summary in easily understandable terms will be included with the final report (Art. 37 ClinO-MD).

2.5 Ethical Conduct of the Investigation

The investigation will be carried out according to the CIP and with principles enunciated in the current version of the Declaration of Helsinki, the European Regulation on medical devices 2017/745 (MDR), the Norms ISO14155 and ISO14971, the ICH-guidelines of Good Clinical Practice (GCP) as applicable, the Swiss Human Research Act (HRA) and its Ordinances and Swiss regulatory authority's requirements.

2.6 Declaration of interests

All investigators and study personnel will declare any potential conflicts of interest (independence, intellectual, financial, proprietary etc.). No financial interests in the EpiSave system or competing products are permitted.

2.7 Patient Information and Informed Consent

The PI explains to each subject the nature of the investigation, its purpose, the procedures involved, the expected duration, the potential risks and benefits and any discomfort it may entail. Each subject is informed that the participation in the investigation is voluntary and that he/she may withdraw from the investigation at any time and that withdrawal of consent will not affect his/her subsequent medical assistance and treatment. The subjects are informed that he/she can ask any question, and consult with family members, friends, their treating physicians or other experts before deciding about their participation in the investigation. Enough time is given to the subjects.

The subjects are informed that authorised individuals other than their treating physician may examine his/her medical records.

All subjects are given a subject information sheet and a consent form describing the investigation and providing sufficient information for the subjects to make an informed decision about their participation in the investigation.

The formal consent of a subject, using the consent form approved by the CEC, is obtained before the subject is submitted to any study procedure.

The subject should read, understand, and voluntarily agree before signing and dating the informed consent form and is given a copy of the signed document. The consent form is signed and dated by the subject and the PI (or her/his designee, when applicable).

The subjects are informed, with their approval, that their personal physicians are informed about the subjects' participation in the clinical investigation, when applicable.

The signed consent form is retained as part of the clinical investigation documents. The subject's medical records are clearly marked to indicate that the subject is enrolled in the clinical investigation.

The subject is provided with well-defined procedures for possible emergency situations related to the performance study. The PI makes the necessary arrangements for emergency treatment.

All participants, including vulnerable individuals such as children and those with cognitive impairments, receive an information sheet and a consent form that clearly describe the study and provide sufficient details to support an informed decision regarding participation. For minors, consent is obtained from their legal guardians, and assent is sought from the child when appropriate. For participants with impaired decision-making capacity, consent is obtained from a legal representative.

The consent process includes simplified materials and additional support to ensure participant understanding. Consent or assent must be given voluntarily and is documented by the signature and date of the participant, their legal representative or parent, and the principal investigator or designated study staff. A copy of the signed consent form is provided to the participant or their representative, while the original is retained in the study records.

Participants who do not wish to take part in the study are not enrolled. Additional safeguards are implemented for vulnerable populations, including simplified consent procedures, enhanced monitoring, and regular reassessment of consent or assent throughout the study. Participants retain the right to withdraw from the study at any time and may request information regarding the progress and results of the study.

2.8 Subject privacy and confidentiality

The Sponsor and the PI affirm and uphold the principle of the subjects' right to privacy and that they shall comply with applicable privacy laws. Especially, anonymity of the subjects shall be guaranteed when presenting the data at scientific meetings or publishing them in scientific journals.

The investigator has appropriate knowledge and skills in the areas of data security and data protection or is able to ensure compliance by calling in appropriate expertise (Art. 5, ClinO-MD).

Individual subject medical information obtained as a result of this investigation is considered confidential and disclosure to third parties is prohibited.

Data Management and Anonymization Protocol

All raw data from the smartwatches shall be handled with utmost discretion by CHUV and will only be accessible to authorized study personnel at CHUV. To ensure participant privacy and confidentiality, a strict data separation protocol is implemented:

- **Local site data management:** A password-protected Excel-based log, called "subject information", will be prepared and maintained at each investigational site. This log contains patient names, dates of birth, and attributes the subject ID code using the formula: study ID + centre acronym + patient number (e.g., EPA-CHV-001 for CHUV, EPA-HUG-001 for HUG, etc.). This Excel log remains at each local site and is not transmitted to CHUV.
- **Data transmission to CHUV:** CHUV will only receive anonymized patient codes (e.g., EPA-CHV-001, EPA-HUG-001) along with the corresponding raw smartwatch data. No sensitive patient details (names, dates of birth, or other identifying information) will be transmitted to CHUV.
- **Access restrictions:** Only the investigators at each site will have access to the identities of the patients enrolled at their respective centres. Access to subject ID codes and the coding key, both during and after the study, will be restricted to locally involved study personnel at each site. CHUV personnel will not have access to the coding key or any patient-identifying information.
- **Code breaking:** The code may only be broken if it is necessary to avert an immediate risk to health of the person concerned, to guarantee the rights of the person (e.g., in revoking the consent), or if a legal basis exists for breaking the code. In such cases, only the local site investigators who maintain the coding key will be involved in the code-breaking process.

On all Case Report Forms (CRFs) and other project-specific documents, participants are identified exclusively by their unique participant number. Access to the protocol is reserved to the project partners and other individuals who are directly implicated in the investigation.

Monitoring, Audits, and Regulatory Inspections

The Principal Investigator and participating institutions shall permit investigation-related monitoring, audits, review by Institutional Review Boards (IRBs)/Independent Ethics Committees (IECs), and regulatory inspections. To support these activities, direct access to all relevant source data and documents

(including medical records and informed consent forms) will be granted to authorized personnel. This access will be provided in compliance with applicable data protection regulations, including the Swiss Federal Act on Data Protection (FADP) and the EU General Data Protection Regulation (GDPR).

For data verification purposes, authorised representatives of the Sponsor, the CA or a CEC may require direct access to parts of the medical records relevant to the investigation, including subjects' medical history.

2.9 Early termination of the investigation

The Sponsor may terminate the investigation prematurely according to certain circumstances, for example:

- ethical concerns,
- insufficient subject recruitment,
- when the safety of the subjects is doubtful or at risk, respectively,
- alterations in accepted clinical practice that make the continuation of the investigation unwise,
- early evidence of benefit or harm of the experimental intervention.

2.10 Clinical investigation plan amendments

Amendments to the CIP may only be initiated and approved by the Sponsor. Suggestions may be submitted by investigators, clinical experts, or regulatory authorities, but require Sponsor approval before implementation.

Substantial amendments (e.g., changes to eligibility criteria, outcomes, analyses, or safety procedures) will be promptly communicated to all relevant parties, including: investigators, Competent Authorities, ethics Committees (CEC/IRB/IEC), study participants (if applicable), Clinical trial registries, Scientific journals (if previously published), other stakeholders directly involved.

All amendments will be documented, dated, and version-controlled, and the latest CIP version will be made available to study personnel. Updated consent forms will be provided and re-consent obtained if required by regulations.

Substantial amendments are only implemented after approval by the CEC (Art. 15 ClinO-MD) and, for category C investigations, after approval by the CA also (Art. 20 ClinO-MD). The use of waivers from the CIP is prohibited (Annex XV, Chapter 2, Art. 3.10 MDR).

Under emergency circumstances, deviations from the CIP to protect the rights, safety and well-being of the subjects may proceed without prior approval by the Sponsor and the CEC (and for category C investigations without prior approval by the CA). Such deviations shall be documented and reported to the Sponsor and the CEC (and to the CA for category C investigations) within 2 days (Art. 34 ClinO-MD).

All non-substantial amendments are communicated to the CEC annually together with the safety report / general progress report of the clinical investigation (Art. 15 ClinO-MD), and for category C clinical investigations to the CA as soon as possible (Art. 20 ClinO-MD). The report will include any deviations from the CIP that may have affected the rights, safety or well-being of the subject or the scientific integrity of the investigation (ISO14155).

2.11 Deviation from the Clinical Investigation Plan

The use of waivers from the CIP is prohibited (Annex XV, Chapter 2, Art. 3.10 MDR).

3. BACKGROUND AND RATIONALE

3.1 Background and Rationale for the clinical investigation

Clinical Need and Medical Problem

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures, affecting approximately 50 million people worldwide (1). According to the International League Against Epilepsy (ILAE), epilepsy is defined as a disease of the brain characterized by an enduring predisposition to generate epileptic seizures and by the neurobiologic, cognitive, psychological, and social consequences of this condition (1).

Among the various seizure types, generalized convulsive seizures (GCS), including generalized tonic-clonic seizures (GTCS) and focal to bilateral tonic-clonic seizures (FBTCS), represent one of the most severe manifestations of epilepsy. These seizures are associated with a 15-20 times higher risk of sudden unexpected death in epilepsy (SUDEP) compared to other seizure types (2). Physical injury occurs in 30-50% of GCS episodes, including falls, fractures, and head trauma. The significant impact on quality of life encompasses fear of seizures, social stigma, and activity restrictions. The economic burden is substantial, with direct and indirect costs estimated at \$15.5 billion annually in the United States alone (9).

Drug-resistant epilepsy affects approximately 30% of individuals with epilepsy, defined as failure to achieve sustained seizure freedom despite adequate trials of at least two appropriately chosen and tolerated anti-seizure medications (ASMs). This population faces particularly high risks, with SUDEP rates of 1.1-5.9 per 1,000 person-years, compared to 0.1-0.3 per 1,000 person-years in the general epilepsy population (2,3). SUDEP affects approximately 60,000 people with epilepsy worldwide each year, with an incidence of 1.2/1,000 patient-years (3,4). SUDEP often affects young adults, children, and adolescents (5,6), particularly those with developmental and epileptic encephalopathies (DEE), where the incidence can reach 6/1,000 patient-years in conditions such as Dravet syndrome (7,8).

Nocturnal Seizures and SUDEP Risk

Nocturnal GCS represent a critical clinical challenge due to their association with SUDEP, which accounts for 8-17% of all deaths in people with epilepsy (2). The term "unexpected" reflects the fact that most patients and their families are unaware of SUDEP risk, as they are rarely informed by their physicians (10). SUDEP victims are typically found deceased in bed by family members or caregivers (11,12), who often report profound and lasting grief, guilt, and trauma. Many families believe that a reliable GCS detection device could have saved their loved ones.

Most SUDEP cases are precipitated by a GCS followed by a fatal post-ictal respiratory arrest, frequently exacerbated by the prone position (12). These events usually occur at night, while the individual is sleeping alone, limiting the chance for timely intervention (12,13). Established risk factors include the presence, frequency and nocturnal occurrence of GCS, living or sleeping alone, and non-adherence to antiseizure medication (13-17). Recently, additional risk factors have been identified, such as peri-ictal apnea (16,17), extratemporal epileptogenic zones (particularly perisylvian) (19), high body mass index (18-21), and lower polygenic risk scores for longevity and intelligence (21).

Clinical evidence demonstrates that nocturnal GCS are associated with more severe hypoxemia (oxygen saturation <70% in 50% of cases), increased postictal generalized EEG suppression (PGES), prolonged postictal immobility periods, and higher risk of cardiac arrhythmias and respiratory depression (12). The MORTEMUS study, the most cited SUDEP study, provided the first description of the sequence of events leading to SUDEP following a GCS, establishing the critical importance of timely intervention during the post-ictal period (12).

Current Limitations in Seizure Detection

Traditional monitoring approaches face significant limitations that compromise effective seizure detection and SUDEP prevention. Patient self-reporting is unreliable, with studies showing that 85% of nocturnal seizures go unreported (36), while caregiver observation is limited by sleep patterns and physical proximity. Inpatient video-electroencephalography (v-EEG) monitoring, while providing gold standard data, is expensive, short-term, and not representative of the home environment where most SUDEP events occur (12). Seizure diaries are subject to recall bias and underreporting, further limiting their clinical utility.

The 2020 practice guideline of the ILAE and the International Federation of Clinical Neurophysiology (IFCN) concluded that there is high-level evidence supporting the accuracy of automated GCS detection and recommends the use of clinically validated wearable devices in selected patients (34). However, currently available medically certified GCS detection devices face several critical limitations that compromise their widespread adoption and effectiveness.

Four medically certified GCS detection devices are currently available: EpiMonitor (Empatica, wrist-worn, 3D-Acc and EDA), Epi-Care Mobile/Free (Danish Care Technology, wrist-worn, 3D-Acc only), Night-Watch (LivAssured, upper arm at night, 3D-Acc and PPG), and EpiWatch (App on the Apple Watch, 3D-Acc and PPG) (35,42-47). While these devices demonstrate varying levels of sensitivity (65-100%) and false alarm rates (FAR) ranging from 0.05-0.7 per day (42-47), they suffer from several critical limitations: high FAR contributing to alarm fatigue and device abandonment, stigmatization and usability issues from wearing dedicated devices, limited functionality including the inability to cancel false alarms, and restricted access and affordability, particularly outside Europe and the United States.

Cost represents a particularly critical barrier to adoption. Most health systems do not reimburse GCS detectors, and many people with drug-resistant epilepsy have limited financial resources. A recent US survey identified cost as the primary reason for non-use of GCS detectors (35). Current pricing ranges from \$250-1,500 USD plus monthly fees for most devices, with only EpiWatch available at device cost only but limited to the Apple ecosystem. These limitations result in poor adoption rates, with only 5-10% of eligible patients using seizure detection devices.

Scientific Background and EpiSave Development

The EpiSave system was developed following a standardized framework for seizure detection device validation (41), building upon two Swiss National Science Foundation (SNSF)-funded grants (#320030_179240 and #CRSII5_193813) that enabled the creation of the world's largest dataset combining v-EEG and wrist-worn sensor recordings in people with epilepsy (55). This dataset, collected through a prospective multi-center study across eight European epilepsy monitoring units, included data from 384 patients and over 600 seizures—104 of which were GCS (55). The same dataset was used for both algorithm development and Phase 2 clinical validation.

The development process involved: (1) algorithm development based on the SNSF-funded projects database; (2) technical innovation through ensemble-based convolutional neural network architecture with tunable sensitivity; and (3) Phase 2 validation using the same cohort. The algorithm was successfully embedded into a low-cost off-the-shelf Android-based smartwatch (TicWatch Pro) with the assistance of the Embedded System Laboratory at EPFL. A professional-grade companion app was developed to implement key functionalities including user account management, caregiver notifications, false alarm cancellation, and secure data streaming.

For Phase 2 clinical validation, the dataset from 384 patients undergoing v-EEG monitoring with wrist-worn three-dimensional (3D) accelerometer sensors was split into training and test sets (55). The training set comprised 37 patients with 54 GCS across 4,922 hours (~206 days) of recording, while the independent test set included 347 patients across 31,479 hours (~1,312 days) of recording, including 33 patients with 49 GCS (55).

Performance metrics on the independent test set demonstrated 96% sensitivity (95% CI: 90%-100%), with a FAR of <1/8 days (95% CI: 1/9-1/7 days) and 1 false alarm per 61 nights (55). Detection latency was a median of 26 seconds after GCS onset. The algorithm's tunable sensitivity allows achievement of up to 100% sensitivity or reduction of false alarms to 1/100 days through quantile adjustment (55). The ensemble-based convolutional neural network (CNN) architecture represents a significant advancement in single-sensor convulsive seizure (CS) detection, introducing two key innovations: (1) an ensemble approach providing inherent robustness against individual model variations; and (2) quantile-based aggregation enabling adaptive sensitivity tuning without requiring model retraining, offering clinicians and patients flexibility in balancing detection sensitivity against false alarms (55).

3.2 Identification and description of the Investigational Medical Device

Brand names(s)	EpiSave
Manufacturer(s)	NeuroDigital@NeuroTech Group (algorithm development); UQIDO (software development)

Name or number of the model(s)/ type(s)	EpiSave system (investigational)
Software version(s)	EpiSave v2.0 investigational version
Software algorithms	EpiSave GCS detection algorithm (proprietary ensemble-based convolutional neural network)
Accessories	None

System Components

The investigational medical device is the **EpiSave system**, which consists of three components:

1. **EpiSave GCS detection algorithm:** The core investigational component, a proprietary deep learning algorithm developed by NeuroDigital@NeuroTech Group, designed to detect GCS, including GTCS and FBTCS, using 3D-acc.
2. **EpiSave smartwatch application:** Software application developed by UQIDO, running on Android Wear OS smartwatches, which embeds the EpiSave algorithm and provides real-time seizure detection and alert functionality.
3. **EpiSave smartphone companion application:** Software application developed by UQIDO, running on Android or iOS smartphones, which receives alerts from the smartwatch app and notifies designated caregivers via SMS and push notifications.

Hardware Platform

Within this investigation, the EpiSave system operates on standard, commercially available devices:

- **Smartwatch:** TicWatch Pro 5 (Mobvoi), running Android Wear OS 4.0+
- **Smartphone:** Participant's own Android (8.0+) or iOS (12.0+) compatible smartphone

The TicWatch Pro 5 smartwatch is a CE-marked consumer electronic device (not CE-marked as a medical device). The hardware platform meets comprehensive safety standards including IEC 60601-1 and IEC 60601-1-2 for electromagnetic compatibility, ISO 10993 standards for skin contact biocompatibility, and IEC 62133 battery safety standards.

CE Marking Status and Intended Purpose

The EpiSave system (software and algorithm) is **not CE-marked** as a medical device and is under investigation in this clinical study. The hardware platform (TicWatch Pro 5 smartwatch) is CE-marked as consumer electronic equipment, not for medical use.

Intended purpose of the investigational device: The EpiSave system is intended for real-time detection of GCS and caregiver alerting in people with epilepsy (PWE), with the primary goal of reducing SUDEP risk through timely intervention. The system also enables large-scale data collection for SUDEP biomarker discovery.

Intended Population

The EpiSave system is intended for PWE, specifically those who:

- Experience GCS
- Are at risk of SUDEP
- Are aged 6 years or older (both children and adults)
- Have a designated caregiver available for monitoring

The system is particularly targeted for:

- Patients who live or sleep alone
- Children who may not be able to reposition themselves after a seizure or alert a caregiver independently
- Individuals with drug-resistant epilepsy who experience frequent nocturnal GCS

Off-Label Use

The TicWatch Pro 5 (TWP5) smartwatch is used off-label in this investigation. The device is CE-marked

as consumer electronic equipment, but is being used in this investigation as a platform for investigational medical device software (EpiSave system). The smartwatch will be used in accordance with its standard consumer specifications, with the investigational EpiSave software application installed.

Device Materials in Contact with Body

The TWP5 smartwatch comes into contact with the skin of the wrist. The device materials comply with ISO 10993 standards for skin contact biocompatibility. The smartwatch does not contain any medicinal substances, human or animal tissues or their derivatives, or other biological active substances. The device complies with applicable national regulations for consumer electronic equipment.

Training and Experience Requirements

The EpiSave system does not require specific medical training or prior experience with medical devices, as it operates as software applications on standard consumer smartwatch and smartphone platforms. Users need basic familiarity with smartphone and smartwatch operation.

Participant training: Research staff will provide comprehensive training on EpiSave system operation at the device setup visit, including:

- Smartwatch and smartphone pairing
- Application installation and configuration
- Caregiver alert system setup
- False alarm reporting procedures
- Daily device charging and maintenance

A user manual for the smartwatch and smartphone applications will be provided to all participants. The EpiSave applications will be installed and configured by the research team on the participant's smartphone and paired with the study-provided smartwatch at the time of inclusion.

Healthcare provider requirements: No specific medical or surgical procedures are involved in the use of the EpiSave system. Healthcare providers involved in the investigation should be familiar with epilepsy management and seizure detection technologies, as detailed in the Investigator's Brochure.

References

For detailed technical specifications, safety information, and operational procedures, refer to:

- **Investigator's Brochure (IB):** Contains comprehensive information on the EpiSave system, including algorithm specifications, performance characteristics, and safety data.
- **Instructions for Use (IFU):** User manual provided to participants, detailing device operation, maintenance, and troubleshooting procedures.

3.3 Preclinical Evidence

Not applicable.

3.4 Clinical Evidence to Date

Available Clinical Experience – Phase 2 Clinical Validation Study

The EpiSave algorithm underwent Phase 2 clinical validation through a prospective, multi-center, observational study conducted across eight European epilepsy monitoring units (55). This study represents the most comprehensive validation of a seizure detection algorithm to date, utilizing the world's largest dataset combining video-EEG and wrist-worn sensor recordings in people with epilepsy.

Study Design and Population: The study population comprised 384 patients (age range: 6-80 years) with focal, generalized, and combined epilepsies, with particular focus on GCS detection (55). Data collection utilized 3D accelerometer and gyroscope sensors at 50Hz sampling rate, with video-EEG monitoring serving as the gold standard for expert annotation (55).

Performance Results: The dataset was split into training (37 patients, 54 GCS, 4,922 hours) and independent test sets (347 patients, 49 GCS, 31,479 hours) (55). Performance metrics on the independent test set demonstrated:

- Sensitivity: 96% (95% CI: 90%-100%)

- False alarm rate: <1/8 days (95% CI: 1/9-1/7 days), equivalent to 1 false alarm per 61 nights
- Detection latency: median 26 seconds (IQR: 15-45 seconds) after GCS onset (55)

Smartwatch Platform Validation: The EpiSave algorithm was subsequently validated on smartwatch hardware through the Enigma clinical study, where the algorithm operated in offline mode (without real-time detection or caregiver notification) to evaluate performance directly on the smartwatch platform. This validation confirmed the algorithm's functionality when embedded in smartwatch hardware.

Current Clinical Experience Status: The EpiSave algorithm has been validated in controlled EMU settings. However, the complete EpiSave system (comprising the algorithm integrated with the smartwatch application, smartphone companion application, and caregiver alert system) has not yet been tested in patients. This investigation represents the first clinical evaluation of the complete EpiSave system in real-world ambulatory settings.

Comparison with Existing Devices:

Devices	Sensitivity (%)	False Alarm Rate (FAR)	Number of patients	Total recording time (days)
Investigational Medical Device				
EpiSave	96	<1/8 days	384	1 312
Existing devices				
EpiMonitor (Empatica)	95-98	Variable FAR	135	120
NightWatch (LivAssured)	80-100	0.7/night	28	90
EpiCare Mobile (DanishCare)	65-90	Variable FAR	15	30
EpiWatch	100	0.05/ day	85	178

EpiSave demonstrates key advantages compared to existing devices: higher sensitivity (96% vs. 65-100% range for other devices), lower false alarm rate (<1/8 days vs. 0.05-0.7 per day), larger validation dataset (384 patients vs. 15-178 patients), longer study duration (1,312 days vs. 30-178 days), and tunable sensitivity allowing adjustment based on patient needs (5)

Adverse Device Effects

No adverse device effects have been reported in the Phase 2 clinical validation study or the Enigma smartwatch validation study. The algorithm operates in offline mode during validation studies, processing accelerometry data without real-time intervention, which minimizes potential risks. No device-related adverse events, serious adverse events, or device deficiencies were documented in these studies.

Device Modifications and Recalls

No device modifications or recalls have been implemented for the EpiSave system. The current investigational version (EpiSave v2.0) represents the first complete system integration of the algorithm with the smartwatch and smartphone applications.

Post-Marketing Experience

Not applicable. The EpiSave system is not yet marketed and has not received regulatory approval for clinical use. This investigation represents the first clinical evaluation of the complete system in real-world settings.

Real-World Performance Considerations

The Phase 2 validation was conducted in controlled EMU environments, which differ from real-world home settings in several critical aspects. EMU settings provide controlled environments with continuous video monitoring and expert staff, but represent artificial settings with limited movement and potential stress factors that may not reflect real-world usage patterns. Home environments offer natural settings with typical daily activities, but present challenges including environmental noise, device wear compliance issues, and variable caregiver availability.

Real-world ambulatory validation is essential to assess: (1) clinical utility of device performance in actual use conditions; (2) user acceptance and device usability; (3) patient satisfaction; and (4) safety monitoring of device-related adverse events in home settings. This investigation addresses the critical gap between controlled clinical validation and real-world clinical implementation.

3.5 Justification for the design of the clinical investigation

Justification for Use of the Medical Device

Clinical Need and Indication: As established in section 3.1, GCS represent one of the most severe manifestations of epilepsy, associated with a 15-20 times higher risk of SUDEP compared to other seizure types (2). Drug-resistant epilepsy affects approximately 30% of individuals with epilepsy, with SUDEP rates of 1.1-5.9 per 1,000 person-years (2,3). Nocturnal GCS represent a critical clinical challenge, as 70-80% of SUDEP cases occur during sleep (2), and most SUDEP victims are found deceased in bed by family members (11,12). The 2020 ILAE/IFCN practice guideline recommends the use of clinically validated wearable devices for GCS detection in selected patients (34), establishing the clinical need for effective seizure detection technology.

Limitations of Current Solutions: As detailed in section 3.1.3, currently available medically certified GCS detection devices face critical limitations: high false alarm rates (0.05-0.7 per day) contributing to alarm fatigue and device abandonment, substantial costs (\$250-1,500 USD plus monthly fees) limiting access, stigmatization from dedicated devices, and restricted availability outside Europe and the United States (35,42-47). These limitations result in poor adoption rates, with only 5-10% of eligible patients using seizure detection devices.

EpiSave System Advantages: Based on Phase 2 clinical validation (section 3.4), the EpiSave algorithm demonstrated 96% sensitivity (95% CI: 90%-100%) with a false alarm rate of <1/8 days, superior to existing devices (55). The system addresses current limitations through: (1) lower false alarm rate reducing alarm fatigue; (2) use of standard consumer smartwatch platform reducing stigmatization; (3) cost-effective approach using off-the-shelf hardware; and (4) tunable sensitivity allowing customization based on patient needs.

Method of Application

Continuous Wear Protocol: The EpiSave system is intended for continuous 24-hour daily wear on the non-dominant wrist, with the device removed only for charging (approximately 2 hours daily). This continuous monitoring approach is justified by: (1) the unpredictable nature of GCS occurrence, requiring constant surveillance; (2) the critical importance of nocturnal detection, as 70-80% of SUDEP cases occur during sleep (2); and (3) the need for immediate caregiver alerting, as timely intervention during the post-ictal period is critical for SUDEP prevention (12).

Smartwatch Platform: The use of a standard consumer smartwatch platform (TicWatch Pro 5) is justified by: (1) reduced stigmatization compared to dedicated medical devices; (2) improved user acceptance and compliance; (3) cost-effectiveness using commercially available hardware; and (4) proven algorithm performance when embedded in smartwatch hardware (Enigma study validation).

Caregiver Alert System: The integrated caregiver alert system, delivering SMS and push notifications upon GCS detection, is justified by: (1) the critical need for timely intervention during the post-ictal period (12); (2) the fact that most SUDEP victims are found by family members or caregivers (11,12); and (3) the delayed caregiver response typically associated with nocturnal seizures, necessitating automated alerting.

Regimen and Period of Intended Use

Investigation Duration: Each participant will use the EpiSave system for 6 months, with total recruitment period of up to 24 months. This duration is justified by: (1) the event-driven design requiring sufficient time to capture ≥75 nocturnal GCS events in ≥50 participants; (2) the need to evaluate device

performance across different seasons and activity patterns; (3) the requirement to assess long-term device wear compliance and user acceptance; and (4) the need to capture sufficient safety data in real-world settings.

Monitoring Period Rationale: The 6-month monitoring period per participant allows for: (1) comprehensive evaluation of device performance across varying seizure frequencies; (2) assessment of seasonal variations in seizure patterns; (3) evaluation of long-term device compliance and user satisfaction; and (4) collection of sufficient data for statistical analysis while maintaining participant engagement.

Study Design Justification

Event-Driven Design: The study employs an event-driven design targeting ≥ 75 nocturnal GCS events in ≥ 50 participants, justified by: (1) statistical power requirements providing 90% power to detect sensitivity $\geq 90\%$ with 95% confidence; (2) clinical relevance focusing on the most clinically significant seizure type (nocturnal GCS); and (3) alignment with standards proposed for phase 4 studies of seizure detection devices in ambulatory patients (72). Anticipated recruitment of 65-80% of patients who will experience a nocturnal GCS during the study period translates to recruitment of 77 to 100 people with epilepsy over 24 months.

Single-Arm, Open-Label Design: This design is appropriate because: (1) medical device validation represents the standard approach for device performance studies; (2) historical controls provide sufficient comparison data from published literature on existing devices; and (3) clinical equipoise exists as there is no established standard of care for seizure detection devices. This design ensures comprehensive evaluation of EpiSave's performance while maintaining scientific rigor.

Ambulatory Setting: Real-world ambulatory validation is essential because: (1) environmental differences between home and EMU settings may significantly affect device performance; (2) natural daily activities and movement patterns may impact detection accuracy; (3) 24-hour daily wear requirements need validation in real-world conditions; and (4) caregiver integration and alert system effectiveness require real-world testing. The study incorporates autonomous camera systems for nocturnal monitoring to generate ground truth for nocturnal GCS detection, with all video recordings stored locally to protect patient privacy.

Nocturnal Focus: The primary focus on nocturnal GCS detection is justified by: (1) 70-80% of SUDEP cases occur during sleep (2); (2) nocturnal GCS are associated with more severe hypoxemia, increased PGES, and prolonged postictal immobility (12); (3) caregiver response is typically delayed for nocturnal seizures; and (4) this represents the highest-risk period for SUDEP occurrence and the most clinically significant application of seizure detection technology.

3.6 Explanation for choice of comparator

This investigation employs a single-arm design with no active comparator or control group. Performance evaluation will be conducted through comparison with historical data from published studies of existing medically certified seizure detection devices.

Scientific Rationale

No Active Control Group: This investigation does not include an active comparator (other medical device, other therapy, sham intervention, or no treatment control) for the following scientific reasons:

1. **Medical Device Validation Standard:** Single-arm designs represent the standard approach for medical device performance validation studies, as established in ISO 14155:2020 and regulatory guidance for medical device investigations.
2. **No Established Standard of Care:** There is no established standard of care for seizure detection devices. As detailed in section 3.1.3, currently available devices face critical limitations (high false alarm rates, cost barriers, limited access), resulting in poor adoption rates of only 5-10% of eligible patients. No single device represents a gold standard comparator.
3. **Historical Control Data Availability:** Sufficient comparison data is available from published literature on existing medically certified devices (EpiMonitor, NightWatch, EpiCare Mobile, Epi-Watch), as detailed in section 3.4. These devices have been evaluated in controlled studies with well-documented performance metrics, providing robust historical controls for comparison.
4. **Primary Objective Focus:** The primary objective is to assess EpiSave system sensitivity for nocturnal GCS detection confirmed by nocturnal home recordings in real-world ambulatory settings. This can be evaluated against a gold standard (nocturnal home video recordings) without

requiring an active comparator, as the investigation aims to establish absolute performance rather than relative superiority.

5. **Regulatory Alignment:** Single-arm designs are appropriate for medical device validation when historical controls provide sufficient comparison data, as recognized in medical device regulatory guidance.

Historical Control Comparison

Performance comparison will be conducted using published data from existing medically certified seizure detection devices, as detailed in section 3.4: EpiMonitor (Empatica), NightWatch (LivAssured), EpiCare Mobile (Danish Care Technology), and EpiWatch (Apple). These devices have been evaluated in published studies with documented performance metrics including sensitivity, false alarm rates, and study populations.

Statistical comparison will employ appropriate methods to account for differences between study populations and designs, ensuring valid comparison while recognizing the limitations inherent in historical control comparisons.

3.7 Risk evaluation (Risk-to-Benefits rationale)

The risk management process for the EpiSave system follows ISO 14971:2019, encompassing risk analysis, risk-to-benefit assessment, and risk control measures. As a non-invasive medical device (Class IIa), the EpiSave system presents inherently low risks compared to invasive devices, with no surgical procedures, implantation, or direct tissue contact beyond standard skin contact with a wrist-worn device.

Anticipated Adverse Device Effects

Based on the clinical evidence to date (section 3.4), no adverse device effects have been reported in Phase 2 clinical validation studies or smartwatch platform validation. The analysis of adverse device effects and device modification history is detailed in section 3.4.

Anticipated adverse device effects (based on device characteristics and similar devices):

1. **Skin irritation or contact dermatitis:** Low risk associated with prolonged skin contact with the smartwatch wristband. The TicWatch Pro 5 uses materials compliant with ISO 10993 standards for skin contact biocompatibility. Risk severity: Minor. Expected frequency: Rare (<1% of participants).
2. **Device-related discomfort:** Low risk of physical discomfort from continuous device wear, including pressure points or interference with sleep. Risk severity: Minor. Expected frequency: Uncommon (1-5% of participants).
3. **False alarms:** Moderate risk of false positive detections leading to unnecessary caregiver alerts. Based on Phase 2 validation, false alarm rate is <1/8 days (section 3.4). Risk severity: Minor to moderate (psychological impact on caregivers). Expected frequency: Common (expected in most participants over 6 months).
4. **Missed seizures (false negatives):** Moderate risk of undetected GCS events. Based on Phase 2 validation, sensitivity is 96% (95% CI: 90%-100%) (section 3.4). Risk severity: Moderate to serious (potential for delayed intervention). Expected frequency: Uncommon (4-10% of GCS events may be missed).

Residual Risks

Residual risks that cannot be completely eliminated despite risk control measures:

1. **System failure leading to missed seizures:** Despite continuous monitoring and backup systems, complete system failure could result in undetected seizures. Residual risk severity: Serious. Mitigation: Comprehensive caregiver training on manual seizure recognition, backup alert systems, and regular device maintenance checks.
2. **Insufficient efficacy:** The device may not achieve the expected performance in real-world settings, potentially providing false reassurance. Residual risk severity: Moderate. Mitigation: Continuous performance monitoring, regular safety assessments, and clear communication that the device is an adjunct to, not replacement for, appropriate supervision.
3. **False negative results:** Missed GCS detection could delay caregiver intervention, potentially

increasing SUDEP risk. Residual risk severity: Serious. Mitigation: Algorithm optimization, continuous performance monitoring, and caregiver education on device limitations.

4. **False positive results:** Frequent false alarms may lead to caregiver fatigue and reduced response effectiveness ("cry wolf" syndrome). Residual risk severity: Moderate. Mitigation: Algorithm optimization to minimize false alarm rate, caregiver training on false alarm management, and false alarm cancellation functionality.

Procedures Involved in Device Use

The EpiSave system is non-invasive and requires no surgical procedures or invasive interventions. Device use involves:

1. **Device application:** Simple placement of the smartwatch on the non-dominant wrist, similar to standard consumer smartwatch use. No invasive procedures required.
2. **Daily maintenance:** Device charging (approximately 2 hours daily), routine cleaning, and basic smartphone app interaction. No medical procedures involved.
3. **Data collection:** Automatic, continuous data collection through the device sensors. No active patient intervention required beyond device wear.
4. **Caregiver response:** Standard caregiver response to seizure alerts, involving standard first aid and intervention procedures already familiar to caregivers of people with epilepsy.

Risks due to interaction with concomitant treatments

Anti-seizure medications (ASMs): No known pharmacological interactions. The EpiSave system does not interfere with ASM metabolism, absorption, or efficacy. Device use does not require modification of existing ASM regimens.

Other medical devices: No known electromagnetic interference with other medical devices. The smartwatch operates using standard Bluetooth and cellular connectivity, which may interfere with certain medical devices in close proximity. Participants using other medical devices will be assessed for compatibility during screening.

Concurrent medical interventions: No contraindications with standard epilepsy care, including vagus nerve stimulation, ketogenic diet, or other epilepsy treatments. The device does not interfere with routine medical care or emergency interventions.

Risks Associated with Participation in the Investigation

Investigation-specific risks:

1. **Time burden:** Participation requires 6 months of device wear and monthly follow-up visits. Risk severity: Minor. Mitigation: Flexible scheduling, remote follow-up options where appropriate.
2. **Privacy concerns:** Continuous monitoring and data collection may raise privacy concerns. Risk severity: Minor. Mitigation: Comprehensive data protection measures (section 2.8), local data storage, and transparent privacy policies.
3. **Device dependency:** Participants may develop psychological dependence on the device. Risk severity: Minor. Mitigation: Clear communication that the device is investigational, post-study transition planning, and psychological support if needed.
4. **Withholding proven therapies:** Not applicable. The investigation does not require withholding any proven therapies. Participants continue their standard epilepsy care, including ASMs and other treatments.

Anticipated Clinical Benefit

Primary clinical benefit: Reliable detection of generalized convulsive seizures (GCS) with 96% sensitivity (section 3.4), enabling timely caregiver intervention during the critical post-ictal period when SUDEP risk is highest. This addresses the critical unmet need identified in section 3.1, where 70-80% of SUDEP cases occur during sleep (2), and most victims are found by family members (11,12).

Secondary clinical benefits:

1. **SUDEP prevention:** Potential to prevent 1-2 SUDEP cases per 1,000 patient-years through timely intervention, addressing the devastating impact of SUDEP affecting approximately 60,000 people with epilepsy worldwide annually (section 3.1).

2. **Injury reduction:** Potential to reduce seizure-related injuries by 30-50% through timely care-giver intervention, addressing the 30-50% of GCS episodes associated with physical injury (section 3.1).
3. **Quality of life improvement:** Reduced seizure-related anxiety and improved independence for patients, with reduced psychological burden for caregivers. Expected improvements of 20-30% in QOLIE-31 scores.
4. **Clinical decision support:** Objective seizure detection data enabling healthcare providers to optimize ASM regimens and assess treatment response more accurately than traditional seizure diaries, addressing the limitations of patient self-reporting where 85% of nocturnal seizures go unreported (section 3.1.3).
5. **Economic benefits:** Potential healthcare cost savings through reduced emergency department visits, hospitalizations, and seizure-related injuries.

Benefit for future subjects: This investigation will generate essential data on real-world performance and safety of the EpiSave system, potentially leading to regulatory approval and wider availability of effective, affordable seizure detection technology. The investigation addresses critical limitations of existing devices (section 3.1.3), potentially improving treatment options for the 30% of people with epilepsy who have drug-resistant epilepsy and are at high SUDEP risk.

Risk-Benefit Rationale

Overall risk-benefit assessment: The risk-benefit ratio is highly favorable. The EpiSave system is a non-invasive device with inherently low risks, while addressing a critical unmet medical need in a high-risk population where current monitoring approaches are inadequate (section 3.1.3).

Risk profile: Characterized by low to moderate severity risks with high mitigability. The most serious risks (missed seizures, system failure) are mitigated through comprehensive risk control measures. The device's non-invasive nature eliminates risks associated with surgical procedures, implantation, or invasive interventions.

Benefit profile: Demonstrates high clinical impact through:

- SUDEP prevention in a population with SUDEP rates of 1.1-5.9 per 1,000 person-years (section 3.1)
- Injury reduction addressing 30-50% of GCS episodes associated with physical injury
- Quality of life improvements for both patients and caregivers
- Economic benefits through healthcare cost savings

Justification: The critical need for effective nocturnal seizure detection in high-risk patients with drug-resistant epilepsy, combined with the comprehensive risk mitigation strategies and the device's non-invasive nature, supports the conclusion that the potential benefits significantly outweigh the identified risks. The system addresses an unmet medical need where the consequences of missed seizures can be fatal, and current monitoring approaches are inadequate.

Risk Control and Mitigation Measures

Risk control measures are implemented according to ISO 14971:2019 principles (reference to risk analysis report: see Investigator's Brochure):

1. **Inherent safety by design:** Non-invasive design eliminates risks associated with invasive procedures. Algorithm optimization minimizes false alarm rate (<1/8 days) and maximizes sensitivity (96%).
2. **Protective measures:**
 - Continuous system monitoring and performance tracking
 - Backup alert systems (SMS and push notifications)
 - False alarm cancellation functionality
 - Regular device maintenance and software updates
3. **Information for safety:**
 - Comprehensive caregiver training on device operation, false alarm management, and manual seizure recognition
 - Clear instructions emphasizing that EpiSave is an adjunct to, not replacement for, appropriate supervision

- User manual and training materials provided to all participants
 - Regular safety assessments and follow-up visits
4. **Training requirements:** All participants and caregivers receive comprehensive training at device setup visit, including device operation, alert response procedures, false alarm management, and manual seizure recognition skills.
 5. **Monitoring and surveillance:** Continuous monitoring of device performance, adverse events, and device deficiencies throughout the investigation period, with immediate reporting of serious adverse events.

Post-Investigation Care

Participants will have the opportunity to continue using the EpiSave system after discharge from the investigation. This continuation ensures continued access to seizure detection technology for participants who benefit from the system, while allowing continued anonymous data collection for research purposes. The research team will facilitate device continuation for interested participants.

3.8 Justification of the choice of the investigation population

Choice of Investigation Population and Rationale

Target Population Definition

The investigation targets people with epilepsy (PWE) who have drug-resistant epilepsy and experience generalized convulsive seizures (GCS), specifically those with frequent nocturnal GCS. This population selection is justified by multiple scientific and clinical considerations:

1. **High SUDEP Risk:** Patients with drug-resistant epilepsy face markedly elevated SUDEP risk (1.1-5.9 per 1,000 person-years) compared to the general epilepsy population (0.1-0.3 per 1,000 person-years) (section 3.1). This represents the population with the highest unmet clinical need where seizure detection technology can provide the greatest benefit.
2. **GCS Association with SUDEP:** GCS, including generalized tonic-clonic seizures (GTCS) and focal to bilateral tonic-clonic seizures (FBTCS), are associated with a 15-20 times higher risk of SUDEP compared to other seizure types (section 3.1). Approximately 90% of SUDEP cases are associated with GCS, making this seizure type the primary target for detection technology.
3. **Nocturnal GCS Focus:** The requirement for ≥ 2 nocturnal GCS per year addresses the highest-risk period for SUDEP, as 70-80% of SUDEP cases occur during sleep (section 3.1). Nocturnal GCS are associated with more severe hypoxemia, increased postictal generalized EEG suppression, and prolonged postictal immobility periods, making timely intervention particularly critical.
4. **Device Performance Optimization:** GCS present distinct motor patterns suitable for accelerometer-based detection, optimizing device performance evaluation. The requirement for ≥ 2 GCS during the last 12 months ensures adequate event frequency for statistical power while maintaining clinical relevance.
5. **Clinical Indication Alignment:** This population aligns with the intended clinical indication of the EpiSave system, which is designed for people with epilepsy experiencing GCS who are at risk of SUDEP (section 3.2).

Representativeness in Relation to Target Population

Target Population for Clinical Use

The target population for clinical use of the EpiSave system encompasses people with epilepsy (aged 6 years and older) who experience GCS and are at risk of SUDEP, including both pediatric and adult populations, both sexes/genders, and individuals with varying levels of cognitive function. This target population reflects the broad clinical need for effective seizure detection technology across diverse patient groups.

Investigation Population Representativeness

The investigation population is designed to be representative of the target population while ensuring scientific validity and safety. The inclusion criteria (section 7.1) ensure that:

1. **Age Representation:** The age range of ≥ 6 years encompasses both pediatric (6-17 years) and

adult (≥ 18 years) populations, ensuring representation across developmental stages and seizure patterns. This age range reflects the intended clinical use population while accounting for device compatibility (wrist size) and cognitive ability for study participation.

2. **Sex and Gender Representation:** The investigation does not exclude participants based on sex or gender. Both male and female participants will be enrolled, ensuring that device performance can be evaluated across both sexes.
3. **Epilepsy Type Representation:** The investigation includes participants with focal, generalized, and combined epilepsies, as long as they experience nocturnal GCS. This ensures that device performance is evaluated across different epilepsy etiologies, reflecting the diversity of the target population.
4. **Geographic and Socioeconomic Representation:** Recruitment through multiple specialized epilepsy centers across Switzerland and other European countries ensures geographic diversity. The use of standard consumer smartwatch technology (rather than expensive dedicated devices) reduces socioeconomic barriers to participation.

Vulnerable Subjects

Rationale for Inclusion of Vulnerable Subjects

This investigation includes two groups of vulnerable subjects: (1) pediatric participants (aged 6-17 years), and (2) participants with cognitive limitations or impaired decision-making capacity. The inclusion of vulnerable subjects is ethically and scientifically justified according to Art. 11 HRA, as comparable results cannot be obtained exclusively from adults capable of judgment.

Pediatric Participants (6-17 years)

Rationale for Inclusion: Pediatric inclusion is essential because:

1. **Unique Clinical Characteristics:** Children and adolescents experience unique seizure patterns, SUDEP risk profiles, and developmental considerations that differ from adults. Device performance in pediatric populations cannot be reliably extrapolated from adult data.
2. **Higher SUDEP Risk:** Children and adolescents with drug-resistant epilepsy, particularly those with developmental and epileptic encephalopathies (DEE), face elevated SUDEP risk, with incidence reaching 6/1,000 patient-years in conditions such as Dravet syndrome (section 3.1). This represents a critical unmet need where seizure detection technology can provide life-saving benefit.
3. **Developmental Considerations:** Pediatric participants may have different device wear patterns, activity levels, and caregiver dynamics compared to adults, requiring direct evaluation in this population.
4. **Long-term Benefit:** Early intervention with effective seizure detection technology in pediatric populations may reduce long-term complications and improve developmental outcomes.
5. **Inability to Self-Advocate:** Children may not be able to alert caregivers or reposition themselves after seizures, making automated detection particularly critical for this population.

Participants with Cognitive Limitations

Rationale for Inclusion: Inclusion of participants with cognitive limitations or impaired decision-making capacity is essential because:

1. **High SUDEP Risk:** Individuals with cognitive limitations, particularly those with developmental and epileptic encephalopathies, face markedly elevated SUDEP risk (section 3.1). This population represents a critical unmet need.
2. **Inability to Self-Advocate:** Participants with cognitive limitations may not be able to alert caregivers or reposition themselves after seizures, making automated detection particularly critical.
3. **Unique Caregiver Dynamics:** Caregiver response patterns and device operation may differ for participants with cognitive limitations, requiring direct evaluation.
4. **Comparable Results Cannot Be Obtained:** Device performance and caregiver response patterns in this population cannot be reliably extrapolated from participants without cognitive limitations.

Protection of Vulnerable Subjects

All vulnerable subjects will be protected through:

- Appropriate consent procedures (parental/guardian consent for minors, legal representative

consent for participants with impaired decision-making capacity)

- Assent procedures for pediatric participants when developmentally appropriate
- Enhanced monitoring and safety assessments
- Regular reassessment of consent/assent throughout the investigation
- Additional safeguards as detailed in section 2.7 (Patient Information and Informed Consent)

Sample Size Considerations for Vulnerable and Non-Vulnerable Subjects

As detailed in section 11 (Statistical Methods), the investigation employs an event-driven design targeting ≥ 75 nocturnal GCS events in ≥ 50 participants. The sample size calculation does not require separate stratification for vulnerable and non-vulnerable subjects, as the primary endpoint (sensitivity for nocturnal GCS detection) is not expected to differ significantly between these groups based on biological plausibility.

3.9 Patient and public involvement

Patient and public involvement (PPI) has been incorporated into the design of this investigation through discussions with patient partners and relatives of patients who died from SUDEP. During two discussion sessions, questions regarding study procedures were addressed, including the acceptability of wearing smartwatches for seizure detection and the use of nocturnal cameras for home-based monitoring.

Patient Partner Involvement

The first session involved patient partners (people with epilepsy experiencing GCS), and the second session involved relatives of patients who died from SUDEP. During these discussions, both the smartwatch-based seizure detection system and the nocturnal camera monitoring were presented and discussed.

Influence on Study Design

Patient partners and relatives provided valuable feedback that influenced study procedures:

1. **Smartwatch Acceptability:** Wearing a smartwatch for seizure detection was well accepted by patient partners, confirming the acceptability of this approach for the target population.
2. **Nocturnal Camera Monitoring:** Home-based nocturnal camera recordings were accepted by patient partners, with the important specification that recordings must be stored locally (not streamed or transferred) to ensure patient privacy and data security. This feedback directly informed the study design, ensuring that all video recordings are stored locally on memory sticks within the cameras, with no data transfer or streaming involved.

Role and Objectives

The role of patient partners and relatives was to provide perspectives based on their experiential knowledge of epilepsy and SUDEP, ensuring that study procedures are appropriate and acceptable to patients and caregivers. Their input helped anticipate potential barriers to participation and informed the design of study procedures to maximize patient acceptance and retention.

4. CLINICAL INVESTIGATION OBJECTIVES

4.1 Overall Objective

To assess the real-world performance of the EpiSave seizure detection system in detecting GCS in ambulatory patients with epilepsy, with particular focus on nocturnal GCS detection for SUDEP prevention.

4.2 Primary Objective

The primary objective is to assess EpiSave system sensitivity for nocturnal GCS detection confirmed by nocturnal home recordings in real-world ambulatory settings.

4.3 Secondary Objectives

The secondary objectives are:

1. To evaluate the system's performance in detecting GCS during different time periods of the day (daytime and nighttime)
2. To assess the system's false alarm rate (FAR) in ambulatory settings overall and per different time periods of the day (daytime and nighttime)
3. To evaluate the caregiver response, specifically the delay between the onset of GCS and the caregiver's intervention
4. To determine how accurately GCS and post-ictal immobility durations are calculated
5. To evaluate device usability scores and adverse events
6. To assess changes in quality of life using QOLIE-31 scores (Quality of Life in Epilepsy Inventory)
7. To evaluate system reliability and technical performance

4.4 Safety Objectives

To assess the tolerability and safety of the EpiSave system in real-world ambulatory settings, including device-related adverse events, psychological impact, and system reliability.

5. CLINICAL INVESTIGATION OUTCOMES

5.1 Primary Outcome

The primary outcome is the EpiSave system sensitivity for nocturnal GCS detection confirmed by nocturnal home recordings. It is the proportion of true nocturnal GCS detected by EpiSave out of total verified nocturnal GCS confirmed by home video recordings. The validation method relies on nocturnal home video recordings captured using an autonomous camera system installed in participants' bedrooms. Sensitivity will be calculated as part of the statistical analysis, with results presented alongside 95% confidence intervals. The target performance for the system is set at a minimum sensitivity of 90%.

5.2 Secondary Outcomes

The secondary outcomes are presented in the table below:

Secondary outcomes	Definition	Validation method	Target
False alarm rate (FAR)	Number of false alarms per patient per day overall and different time period (nighttime/daytime)	Patient/caregiver feedback through app interface	<1 false alarm per week
Caregiver response time	Time from GCS onset to caregiver intervention	Reported response times through app interface (seizure event log)	≤5 minutes for 90% of alerts
Seizure duration precision	Accuracy of GCS and post-ictal immobility duration calculations	Comparison with video recordings	±10% accuracy
Device usability score	Standardized usability assessment	SUS questionnaire and custom EpiSave questionnaire	t: ≥4.0/5.0 satisfaction score
Quality of life impact	Change in QOLIE-31 scores	Baseline and 6-month follow-up	Clinically meaningful improvement

5.3 Other Outcomes of Interest

Other outcomes of interest include system uptime maintained at ≥95% throughout the study duration, daily battery charging requirements, and robust performance across a range of environmental conditions and user activities.

5.4 Safety Outcomes

Safety outcomes include device-related and procedure-related adverse events (AE), serious adverse events (SAE) such as life-threatening conditions or hospitalizations, device deficiencies involving technical malfunctions that could impact safety, and biocompatibility issues including skin reactions or allergic responses.

6. CLINICAL INVESTIGATION DESIGN

6.1 General clinical investigation design and justification of design

Design Type and Framework

This investigation employs a **prospective, interventional, multi-center, event-driven, single-arm, open-label** design. The investigation framework is **exploratory** for device performance validation, with the primary objective of establishing absolute performance metrics (sensitivity, false alarm rate) rather than demonstrating superiority, equivalence, or non-inferiority relative to a comparator.

Design Rationale

The single-arm, open-label design is scientifically justified for the following reasons:

1. **Medical Device Validation Standard:** Single-arm designs represent the standard approach for medical device performance validation studies, as established in ISO 14155:2020 and regulatory guidance for medical device investigations (section 3.7).
2. **Primary Objective:** The primary objective is to assess EpiSave system sensitivity for nocturnal GCS detection confirmed by nocturnal home recordings, not to demonstrate relative superiority to a comparator. This absolute performance assessment does not require a control group.
3. **Historical Controls:** Sufficient comparison data is available from published literature on existing medically certified seizure detection devices (section 3.4), providing robust historical controls for performance comparison.
4. **Real-World Validation:** The investigation aims to validate device performance in real-world ambulatory settings, where naturalistic conditions are essential for clinical relevance. A control group would not provide additional scientific value for this objective.
5. **Ethical Considerations:** As detailed in section 3.7, providing all participants with access to the investigational device is ethically justified given the high unmet clinical need and the absence of an established standard of care.

Intervention to be Studied

Medical Device: The EpiSave system, comprising:

- EpiSave GCS detection algorithm (proprietary ensemble-based convolutional neural network)
- EpiSave smartwatch application (Android Wear OS)
- EpiSave smartphone companion application (Android/iOS)
- TicWatch Pro 5 smartwatch hardware platform

Procedures:

- Continuous device wear on the non-dominant wrist for approximately 22-24 hours daily (with 2 hours for charging)
- Autonomous nocturnal camera monitoring (Tapo camera system) for ground truth establishment
- Caregiver alert system activation upon GCS detection
- Monthly follow-up visits (by phone) at Weeks 4, 8, 16, 20 for data review, compliance assessment, and safety monitoring
- In-person visit at Week 12 (Month 3) for comprehensive assessment and video retrieval
- End-of-study visit (in-person) at Week 24 for device return, final assessments, questionnaires, and video retrieval

Population to be Studied and Number of Subjects

Population: People with epilepsy (PWE) aged ≥ 6 years with drug-resistant epilepsy experiencing ≥ 2 nocturnal GCS per year, as detailed in section 7.1 (Eligibility criteria).

Sample Size: The investigation employs an **event-driven design** targeting ≥ 75 nocturnal GCS events confirmed by nocturnal home recordings in ≥ 50 participants. Based on anticipated recruitment rates and seizure frequency, this corresponds to an estimated enrollment of 77-100 participants over a 24-month recruitment period. The event-driven approach ensures adequate statistical power (90% power to detect sensitivity $\geq 90\%$ with 95% confidence) while accounting for real-world variability in seizure frequency (section 11.2).

Level and Method of Blinding/Masking

Blinding Status: This is an **open-label investigation**. No blinding or masking procedures are employed.

Kind of Comparator(s)

No Active Comparator: This investigation does not include an active comparator (other medical device, other therapy, sham intervention, or no treatment control), as detailed in section 3.6 (Explanation for choice of comparator).

Historical Controls: Performance comparison will be conducted using published data from existing medically certified seizure detection devices (EpiMonitor, NightWatch, EpiCare Mobile, EpiWatch), as detailed in section 3.4.

Investigation Configuration: Single-arm, parallel design (all participants receive the same intervention simultaneously, no cross-over).

Method of Assignment to Intervention

Randomization: Not applicable. This investigation employs a single-arm design in which all participants receive the same investigational intervention (EpiSave system). No randomization or allocation procedures are required.

Stratification: Not applicable. As all participants receive the same intervention, no stratification is required for treatment assignment.

Assignment Method: Consecutive enrollment of eligible participants who meet inclusion criteria and provide informed consent. Participants are assigned to the investigational intervention upon enrollment, following completion of screening and informed consent procedures (section 7.2).

Expected Duration of Subject Participation

Individual Participation Duration: Each participant will participate in the investigation for **6 months** from device setup to end-of-study visit.

Total Investigation Duration:

- **Recruitment Period:** Up to 24 months
- **Follow-up Period:** 6 months per participant
- **Total Investigation Duration:** Up to 30 months (24 months recruitment + 6 months follow-up for last enrolled participant)

Description of Sequence and Duration of All Investigation Periods

The investigation consists of the following periods:

1. **Screening Period** (Week -2 to Week 0, up to 2 weeks):
 - Medical record review
 - Eligibility assessment
 - Informed consent process
 - Baseline assessments
2. **Device Setup and Training Period** (Week 1, 1 week):
 - Smartwatch and smartphone setup
 - Application installation and configuration
 - Caregiver alert system setup
 - Comprehensive training on device operation
 - Camera system installation and training
3. **Monitoring Period** (Week 2 to Week 24, 22 weeks):
 - Continuous device wear (22-24 hours daily)
 - Continuous nocturnal camera monitoring
 - Monthly follow-up visits (by phone) at Weeks 4, 8, 16, 20
 - In-person visit at Week 12 (Month 3)
 - Real-time seizure detection and caregiver alerting
 - Daily smartwatch data upload (automatically when connected to WiFi with internet connection, or via Bluetooth through paired smartphone when smartwatch is on charger)

- Data collection and compliance monitoring
- 4. **End-of-Study Period** (Week 24, 1 week):
 - Final visit (in-person)
 - Video retrieval for EpiSave events detected and missed GCS reported by caregiver/patient
 - Device return
 - Final assessments and questionnaires
 - Data download and device data deletion
 - Post-investigation care planning
- 5. **Post-Investigation Period** (optional):
 - Participants may continue using the EpiSave system
 - Continued anonymous data collection for research purposes

Known or Potential Problems and Limitations of the Design

1. **Open-Label Design Limitations:**
 - **Potential Reporting Bias:** Participants and caregivers aware of device assignment may exhibit reporting bias (e.g., over-reporting false alarms or under-reporting missed seizures). **Mitigation:** Objective video confirmation provides gold standard validation for nocturnal GCS, independent of participant reporting.
2. **Absence of Active Control Group:**
 - **Limited Comparative Conclusions:** The single-arm design limits direct comparative conclusions relative to other devices. **Mitigation:** Historical control comparison using published data from existing devices provides context for performance evaluation.
3. **Event-Driven Design Considerations:**
 - **Variable Recruitment Timeline:** The event-driven design may result in variable recruitment timelines depending on seizure frequency. **Mitigation:** The design ensures adequate statistical power regardless of recruitment timeline, and recruitment targets (77-100 participants over 24 months) account for variability.
4. **Real-World Setting Challenges:**
 - **Device Compliance:** Real-world device wear compliance may vary, potentially affecting performance evaluation. **Mitigation:** Continuous compliance monitoring, regular follow-up visits, and participant support ensure optimal compliance.
 - **Environmental Variability:** Real-world conditions (activity patterns, environmental factors) may affect device performance differently than controlled settings. **Mitigation:** This variability is intentional and necessary for real-world validation, representing the intended use environment.
5. **Video Monitoring Limitations:**
 - **Privacy Concerns:** Continuous video monitoring may raise privacy concerns. **Mitigation:** Local data storage (no streaming or transfer), clear privacy communication, and option to disable recording address privacy concerns.
 - **Camera Coverage:** Camera coverage may not capture all seizure events, potentially affecting ground truth establishment. **Mitigation:** Cameras are positioned to maximize bedroom coverage, and participants are trained on optimal camera placement.
6. **Caregiver Response Variability:**
 - **Response Time Variability:** Caregiver response times may vary, affecting secondary outcome assessment. **Mitigation:** Automated response time logging and caregiver training ensure consistent response procedures.

Methods for minimising bias

Randomisation: Not Applicable.

Blinding procedures: Investigation Status: This is an **open-label investigation**. No blinding or masking procedures are employed for participants, caregivers, or investigators, as detailed in section 1.1.

Blinded Outcome Assessment: To minimize bias in primary outcome assessment, **blinded video review** is employed:

- **Video Reviewers:** Trained clinicians reviewing video recordings for seizure confirmation are **blinded to device detection status**. Reviewers assess video recordings independently without knowledge of whether EpiSave detected the event, ensuring objective evaluation of device performance.
- **Blinding Procedure:** Video recordings are reviewed in chronological order without device detection information. Device detection status is only linked to video-confirmed events after blinded review is complete, ensuring unbiased assessment of sensitivity and false alarm rate.
- **Blinding Maintenance:** Video reviewers do not have access to device logs or detection information during review. Blinding is maintained until all video reviews are complete and primary outcome analysis is finalized.

Unblinded Parties (as detailed in section 1.1):

- Participants: Aware of device assignment (required for device operation)
- Caregivers: Aware of device assignment (required for alert response)
- Investigators: Aware of device assignment (required for device setup and training)
- Study coordinators: Aware of device assignment (required for study conduct)
- Data analysts: Unblinded (device performance metrics are objective measurements)

Other methods for minimising bias

Several additional measures are implemented to minimize bias:

1. **Standardized Training and Procedures:**
 - All participants receive identical, standardized training on device operation, ensuring consistency in device handling and reducing operator-dependent variability.
 - All investigators follow standardized procedures for device setup, training, and data collection, ensuring consistency across investigational sites.
2. **Objective Measurements:**
 - **Automated Data Collection:** Device performance data (seizure detections, false alarms, response times) are automatically logged by the device, minimizing subjective interpretation and reducing measurement bias.
 - **Video Validation:** Seizure events are confirmed through independent, blinded video analysis, providing objective gold standard validation independent of participant or investigator reporting.
3. **Validated Questionnaires:**
 - Quality of life assessment uses the validated QOLIE-31 questionnaire (Quality of Life in Epilepsy Inventory), ensuring standardized, validated measurement of quality of life outcomes.
 - Device usability assessment uses the standardized System Usability Scale (SUS) questionnaire, ensuring validated measurement of usability outcomes.
4. **Pre-Specified Statistical Analysis Plan:**
 - All analyses are pre-specified in the statistical analysis plan (section 11.4), reducing the risk of data-driven analysis and ensuring objective evaluation of outcomes.
 - Primary and secondary outcomes are clearly defined a priori, with pre-specified analysis methods and success criteria.
5. **Comprehensive Data Collection:**
 - Multiple data sources (device logs, video recordings, participant reports, caregiver reports) provide comprehensive data collection, enabling cross-validation and reducing reliance on single data sources.
6. **Quality Assurance Measures:**
 - Regular monitoring and data quality checks ensure data integrity and completeness.
 - Source data verification procedures ensure accuracy of collected data.

Unblinding Procedures (Code break)

Not Applicable: Unblinding procedures are not applicable to this investigation due to the open-label design. No blinding is employed, and therefore no unblinding procedures are required.

In the event of investigation suspension or premature termination, no unblinding procedures are required, as all parties are already aware of device assignment.

7. CLINICAL INVESTIGATION POPULATION

7.1 Eligibility criteria

Inclusion criteria		Exclusion criteria	
Age requirement			
- Age ≥ 6 years at the time of enrolment		- Age < 6 years at the time of enrolment	
Informed consent			
- Written informed consent from patient or legal representative - For participants aged 6-17 years: assent from the participant and consent from legal guardian		- Unable to provide informed consent or assent and no legal representative available	
Epilepsy diagnosis			
Confirmed diagnosis of epilepsy documented by a board-certified neurologist <i>Diagnosis established through clinical evaluation, EEG findings, and/or neuroimaging as per ILAE diagnostic criteria</i>		- Uncertain or unconfirmed epilepsy diagnosis - History of non-epileptic psychogenic seizures only - Active status epilepticus within the last 3 months	
Drug resistant epilepsy			
- Persisting seizures despite adequate trials of at least two anti-seizure medication (ASM) <i>Adequate ASM trial defined as: appropriate dose for the specific medication, adequate duration (minimum 3 months), and good compliance</i>		- Inadequate ASM trials (<2 adequate ASM trials)	
GCS history and frequency			
- Documented history of GCS (including GTCS or FBTCS) - ≥2 GCS during the last 12 month - ≥2 nocturnal GCS per year		- No GCS risk during the study period - ≤ 2 GCS during the last 12 months - ≤ 2 nocturnal GCS per year	
Device compatibility and Comfort			
- Patient must be comfortable wearing wrist-worn devices - No known hypersensitivity or skin conditions that would prevent smartwatch wear - No physical disabilities that would prevent wearing a wrist-worn device - Willingness to wear the device daily as long as possible (for 22-24 hours per day - with 2 hours for charging)		- Inability or unwillingness to wear wrist-worn devices - Known hypersensitivity to smartwatch material or skin conditions that would prevent smartwatch wear - Physical disabilities that would prevent wearing a wrist-worn device	
Technical Requirements			
- Owing/having access to a compatible smartphone (Android or iOS devices from 2020 and later) - Normal smartphone cellular connectivity for SMS transmission - Ability to charge the smartwatch daily		- No access to compatible smartphone - No cellular connectivity for SMS transmission - Inability to charge the smartwatch daily	
Caregiver Requirements			
- Must have a designated caregiver available for monitoring - Caregiver must be living with the patient or in close proximity (maximum 5 minutes travel		- No designated caregiver available for monitoring - Caregiver not living with patient or not in close proximity	

time) - Caregiver must be able to respond to alerts within XX minutes at night - Caregiver must be willing and able to operate the EpiSave system	- Caregiver unable to respond to alerts within XX minutes at night - Caregiver unwilling or unable to operate the EpiSave system
Medical and safety exclusions	
	- Unstable medical condition that would interfere with study participation - Active psychiatric illness that would impair study compliance

7.2 Recruitment and screening

Recruitment of participants will be consecutive and ongoing throughout the duration of the project. Participants will be recruited either during their EMU hospitalization or at their routine clinical appointment in the outpatient epilepsy clinic. Patients who participated in EpiSave Study 1 during their EMU hospitalization may extend their participation by joining Phase 2 of the application testing directly. Eligible candidates will be invited to participate in the study by the physician locally responsible for the study and/or the study manager and/or the study nurse during an explanatory consent meeting, which might take place prior to or during the EMU hospitalisation or take place during routine clinical appointment. The recruitment period is expected to last 24 months.

The screening process is conducted in three distinct phases to ensure comprehensive evaluation of potential participants:

- Initial screening occurs between **Week -2 and Week 0** and involves thorough review of medical records to confirm epilepsy diagnosis and GCS history, assessment of ASM trial history for drug resistance confirmation, evaluation of seizure frequency over the last 12 months, and assessment of device compatibility and caregiver availability. This comprehensive screening ensures that only appropriate candidates proceed to the informed consent process.
- **The informed consent process** takes place at **Week 0** and includes detailed explanation of study procedures and requirements, comprehensive review of inclusion and exclusion criteria, thorough discussion of risks and benefits associated with participation, and obtainment of written informed consent or assent as appropriate. Subjects must be given sufficient time to consider the information and consult with relatives and experts before providing consent. This process ensures that participants and their families have complete understanding of the investigation before making the decision to participate.
- **Baseline assessment** is conducted at **Week 0** and includes complete medical history and physical examination, documentation of current ASM regimen, assessment of seizure frequency and characteristics, and evaluation of caregiver availability and technical capabilities. This comprehensive baseline assessment provides essential data for the investigation while ensuring that all participants meet the study requirements and can safely participate in the investigation.

7.3 Assignment to investigation groups

Not applicable due to the single-arm design of the study, in which all participants receive the investigational device.

7.4 Not applicable due to the single-arm design of the study, in which all participants receive the investigational device. Criteria for withdrawal / discontinuation of subjects

Participants may be withdrawn from the investigation based on specific criteria designed to ensure participant safety and study integrity. Withdrawal criteria include participant request at any time during the investigation, safety concerns that arise during the course of the study, protocol violations that compromise study integrity, inability to comply with study procedures despite appropriate support, and loss to follow-up that prevents adequate data collection.

Participants who withdraw from the investigation will have their data collected up to the point of withdrawal, ensuring that valuable information is not lost while respecting participant autonomy and safety. This approach maximizes the scientific value of the investigation while maintaining appropriate ethical standards for participant withdrawal and data collection.

8. CLINICAL INVESTIGATION INTERVENTION

8.1 Identity of the medical device under investigation

Experimental Intervention (medical device)

The investigational medical device under evaluation is the EpiSave smartwatch and smartphone application with proprietary GCS detection algorithm. This system represents a novel approach to seizure detection technology, utilizing advanced machine learning algorithms embedded within commercially available smartwatch hardware to provide continuous, real-time monitoring of seizure activity in ambulatory settings.

The hardware platform consists of the TicWatch Pro 5 smartwatch running Android Wear OS 2.0+, which provides the necessary sensor infrastructure for seizure detection. The device specifications include a 3D accelerometer operating at 50Hz sampling rate, gyroscope at 50Hz, PPG sensor for heart rate monitoring, blood oxygen saturation (SpO2) sensor, GPS capability for location services, Bluetooth 4.0+ connectivity for data transmission, 24-hour continuous operation battery life, and IP67 water resistance rating. These specifications ensure robust performance in real-world ambulatory settings while maintaining user comfort and device reliability.

The software components encompass the core EpiSave v2.0 application, which represents the investigational version specifically developed for this clinical investigation. The proprietary machine learning algorithm for GCS detection has been trained on the world's largest dataset combining video-EEG and wrist-worn sensor recordings, enabling accurate detection of generalized convulsive seizures with demonstrated sensitivity of 96% in Phase 2 validation studies. The data collection system provides continuous 24/7 monitoring with enhanced 15-minute post-seizure recording capabilities, while the alert system implements dual-alert functionality to both patients and caregivers via SMS with integrated location information.

The mode of action involves continuous monitoring of motion data through the smartwatch's 3D-accelerometry sensor, which captures the characteristic motor patterns associated with generalized convulsive seizures. Real-time analysis is performed using the proprietary GCS detection algorithm, which processes accelerometry data with less than 2-second detection latency. Upon seizure detection, the system automatically initiates collection of additional biosignals for 15 minutes post-seizure, providing comprehensive documentation of the seizure event and post-ictal period. The automated dual-alert system ensures immediate notification to both patients and caregivers, facilitating timely intervention and emergency response.

Control Intervention (standard/routine/comparator)

This investigation employs a single-arm study design without an active control intervention, which is appropriate for medical device validation studies where the primary objective is to evaluate device performance against established performance standards rather than comparing different interventions. The absence of an active control is justified by the fact that there is no established standard of care for seizure detection devices, and the investigation focuses on demonstrating the safety and effectiveness of the EpiSave system in real-world ambulatory settings.

Historical controls will be utilized for performance comparison, with the study comparing EpiSave performance against published performance metrics from currently certified seizure detection devices, including EpiMonitor (Empatica), NightWatch (LivAssured), EpiCare Mobile (Danish Care Technology), and EpiWatch (Apple). This approach provides meaningful context for evaluating the EpiSave system's performance while maintaining the scientific rigor necessary for regulatory submission.

Labelling and Supply (re-supply)

The EpiSave system utilizes commercial off-the-shelf smartwatches (TicWatch Pro 5) with the investigational EpiSave application pre-installed. Device labeling will comply with investigational device labeling requirements while being appropriate for the consumer electronics platform. The investigational status will be clearly indicated through the EpiSave application interface and accompanying documentation, including appropriate warnings about the investigational nature of the software and usage instructions specific to the study.

The labeling will include device identification, investigational status of the EpiSave application, intended use for seizure detection research, and contact information for technical support and adverse event reporting. Since the hardware platform consists of commercially available smartwatches, the physical labeling will be minimal and focused on identifying the device as part of an investigational study, while the primary investigational labeling will be integrated into the EpiSave application interface.

Supply management will be conducted through a centralized system that tracks the distribution and return of smartwatches with pre-installed investigational software. This system will maintain detailed records of device distribution, usage, and return, ensuring proper inventory control and regulatory compliance. Re-supply procedures will be implemented to replace devices as needed for technical issues, with the ability to quickly install the investigational EpiSave application on replacement devices. The supply chain will leverage standard consumer electronics distribution channels while maintaining the additional documentation and tracking requirements necessary for investigational device regulations.

Storage Conditions

The EpiSave system utilizes commercial off-the-shelf smartwatches (TicWatch Pro 5) with the investigational EpiSave application pre-installed. Storage conditions will follow standard consumer electronics storage practices, as the hardware platform consists of commercially available smartwatches that are designed for consumer use. Devices will be stored at room temperature in dry conditions to maintain battery health and device functionality.

The investigational component (EpiSave application and algorithm) is software-based and does not require special storage conditions beyond those appropriate for the underlying smartwatch hardware. The application will be pre-installed on devices prior to distribution, and devices will be configured with the investigational settings and parameters required for the study. Storage facilities will maintain appropriate security measures to protect device integrity and prevent unauthorized access to investigational software.

Transport of devices will be conducted using standard shipping methods appropriate for consumer electronics, with appropriate packaging to prevent damage during transit. Handling procedures will follow standard protocols for consumer electronics while maintaining the additional security and documentation requirements necessary for investigational device regulations, including proper labeling of devices as investigational and maintenance of device accountability records.

8.2 Discontinuation or modifications of the intervention

Intervention discontinuation may be necessary under specific circumstances to ensure participant safety and study integrity. Discontinuation criteria include the emergence of safety concerns that compromise participant welfare, device malfunction that affects device function or safety, participant request for withdrawal from the investigation, and study termination due to regulatory or administrative reasons. These criteria are designed to protect participant safety while maintaining the scientific integrity of the investigation.

Any modifications to the investigational device during the course of the study will require protocol amendment and regulatory approval through the appropriate channels. This ensures that any changes to the device are properly evaluated for safety and effectiveness implications and that regulatory authorities are informed of modifications that may affect the investigation's conduct or results.

8.3 Compliance with clinical investigation intervention

Compliance monitoring is essential to ensure that participants are using the EpiSave system as intended and that the investigation can achieve its scientific objectives. Device wear time tracking will monitor adherence to the target of 22 hours per day of device wear, ensuring that participants maintain adequate device usage for effective seizure detection. Data transmission monitoring will verify that the system is functioning properly and that seizure data is being collected and transmitted as expected.

Caregiver response tracking will monitor the effectiveness of the alert system by documenting caregiver response times and intervention patterns. Monthly compliance assessments will be conducted to evaluate overall participant compliance with study procedures and identify any issues that may require intervention or support. These monitoring activities ensure that the investigation maintains high standards of compliance while providing necessary support to participants and caregivers.

8.4 Data Collection and Follow-up for withdrawn subjects

Participants who withdraw from the investigation will have complete data collection conducted up to the point of withdrawal, ensuring that valuable information is not lost while respecting participant autonomy. This approach maximizes the scientific value of the investigation while maintaining appropriate ethical

standards for participant withdrawal. Safety follow-up will be conducted for 30 days post-withdrawal to monitor for any delayed adverse events or safety concerns that may be related to device use.

The medical follow-up of withdrawn subjects, or of subjects that drop out from the investigation prematurely is described in chapter 9.2.5 and chapter 9.2.6.

8.5 Clinical investigation specific preventive measures

The investigation incorporates comprehensive safety measures designed to prevent adverse events and ensure participant safety. Comprehensive caregiver training will be provided to ensure that caregivers understand how to use the EpiSave system effectively and respond appropriately to seizure alerts. Emergency response protocols will be established to guide caregivers in responding to seizure events and managing emergency situations.

Backup alert systems will be implemented to ensure that seizure alerts are delivered even if primary systems fail, providing redundancy for critical safety functions. Regular system monitoring will be conducted to identify and address any technical issues that may affect device performance or safety. These preventive measures ensure that the investigation can proceed safely while providing optimal protection for participants.

8.6 Concomitant Interventions (treatments)

Participants will be allowed to continue their standard anti-seizure medications and routine medical care throughout the investigation, as these treatments are essential for their ongoing epilepsy management and do not interfere with the evaluation of the EpiSave system. However, the use of other seizure detection devices during study participation will be restricted to prevent interference with the evaluation of the EpiSave system's performance and to ensure that the investigation can accurately assess the device's effectiveness.

8.7 Medical Device Accountability

Device tracking will be conducted through a centralized tracking system that monitors all devices throughout the investigation period. This system will maintain detailed records of device distribution, usage, location, and status, ensuring comprehensive device accountability and regulatory compliance. Regular inventory checks and device status monitoring will be conducted to verify device integrity and identify any issues that may require attention.

8.8 Return, Analysis or Destruction of the Medical Device

In case of device deficiencies, including malfunction, usability issues, or inadequacy in the information supplied by the manufacturer including labelling, the devices will be returned to the sponsor for analysis.

All devices will be returned at study completion or upon participant withdrawal, ensuring proper device accountability and data security. Device data will be analysed for study endpoints, providing essential information for evaluating the EpiSave system's performance and safety profile. Devices containing personal data will be subject to secure disposal procedures to ensure that sensitive information is properly protected and that devices are disposed of in accordance with regulatory requirements and data protection standards.

9. CLINICAL INVESTIGATION ASSESSMENTS

9.1 Clinical investigation flow chart(s) / table of clinical investigation procedures and assessments

The clinical investigation follows a structured timeline designed to ensure comprehensive evaluation of the EpiSave system's performance and safety in real-world ambulatory settings.

Investigation Periods	Screening	Follow-up visits each month by phone during monitoring with smartwatch and camera					End of study
Visit number	1	2	3	4	5	6	7
Time (week)	0	4	8	12	16	20	24
Visite duration (minutes)	60	5-10	5-10	5-10	5-10	5-10	60
Written consent (ICF)	x						
In- /Exclusion Criteria	x						
Medical History	x						
Demographics	x						
Smartwatch and surveillance camera provided to participant	x						
Participant and caregiver training on the use of the study equipment	x						
System validation and testing	x						
Confirmation of alert system functionality	x						
QOLIE-31 questionnaire	x						x
Fatigue/Seizure Recovery Questionnaire		After the occurrence of a GCS					
Compliance assessment		x	x	x	x	x	x
Adverse Events, Adverse device effects		x	x	x	x	x	x
Device performance evaluation		x	x	x	x	x	x
Data collection verification		x	x	x	x	x	x
Device Deficiencies		x	x	x	x	x	x
Smartwatch and surveillance camera return							x
Download and deletion of data from the smartwatch							x
SUS questionnaire							x
Custom EpiSave system questionnaire							x

9.2 Assessments of outcomes

Assessment of primary outcome

The primary outcome assessment focuses on EpiSave sensitivity for video-confirmed nocturnal GCS detection, prevention. The assessment methodology utilizes home video recordings with autonomous camera systems installed in participants' bedrooms to provide ground truth for seizure confirmation. This approach ensures that the evaluation occurs in real-world settings where the device will be used clinically, rather than in artificial hospital environments.

The analysis employs blinded video analysis for seizure confirmation, with trained clinicians reviewing video recordings to establish the presence and characteristics of nocturnal GCS events. This blinded approach ensures objective evaluation of the EpiSave system's performance without bias from knowledge of device alerts or detections. The assessment timeline spans the entire 6-month study period, providing comprehensive evaluation of device performance across different seasons, activity patterns, and seizure frequencies.

Assessment of secondary outcomes

Secondary outcome assessments encompass multiple dimensions of device performance and clinical impact. The false alarm rate assessment utilizes patient and caregiver feedback through the EpiSave application interface, allowing participants to report and categorize false alarms directly through the system. This approach provides real-time feedback on device performance while minimizing the burden on participants and caregivers.

Caregiver response time assessment employs automatically logged response times captured by the EpiSave system when caregivers respond to seizure alerts. This automated approach ensures accurate documentation of response times without relying on subjective recall or manual timing. Seizure duration precision assessment compares EpiSave-calculated seizure durations with durations measured from video recordings, providing objective evaluation of the device's ability to accurately characterize seizure events.

Device usability assessment utilizes the standardized System Usability Scale (SUS) questionnaire combined with a custom EpiSave questionnaire to evaluate user satisfaction and device acceptability. Quality of life assessment employs the Quality of Life in Epilepsy (QOLIE-31) questionnaire administered at baseline and 6 months to evaluate the impact of the EpiSave system on participants' quality of life and seizure-related anxiety.

Assessment of other outcomes of interest

Additional outcomes of interest provide comprehensive evaluation of the EpiSave system's technical performance and reliability. System uptime assessment involves continuous monitoring of device operation and data transmission, ensuring that the system maintains reliable function throughout the study period. Battery performance assessment utilizes daily charging logs collected through the application to evaluate device power management and charging requirements.

Environmental robustness assessment employs activity-based performance analysis to evaluate how the EpiSave system performs across different environmental conditions, activity levels, and daily routines. This assessment ensures that the device maintains reliable performance in the diverse conditions encountered in real-world ambulatory settings.

Assessment of safety outcomes

Safety outcome assessment encompasses comprehensive monitoring of adverse events, serious adverse events, and device deficiencies throughout the investigation period. Adverse event monitoring involves continuous surveillance through the EpiSave system combined with monthly clinical assessments to identify and document any device-related or study-related adverse events. Serious adverse events receive immediate reporting and follow-up according to regulatory requirements, ensuring prompt evaluation and appropriate intervention.

Device deficiency assessment involves systematic documentation of technical issues, malfunctions, or performance problems that may affect device safety or effectiveness. This assessment ensures that any technical issues are promptly identified, documented, and addressed to maintain participant safety and study integrity.

9.2.1.1 Adverse events

All adverse events (AEs) occurring during the investigation will be documented and assessed according to standard clinical research procedures. The following information will be collected for each AE:

- Time of onset
- Duration
- Resolution status
- Actions taken
- Assessment of intensity (mild, moderate, severe)
- Relationship to the medical device (smartwatch)
- Relationship to study procedures
- Expectedness
- Seriousness

Given the nature of the medical device (the EpiSave app on a commercial smartwatch), the risk of adverse events is minimal. Potential AEs may include, but are not limited to, skin irritation, discomfort due to prolonged wear, or technical issues with the device.

At each scheduled study visit, the investigator will actively ask the participant (and/or caregiver) about any adverse events since the last contact. In addition, participants and caregivers will be encouraged to report any spontaneous adverse events at any time during the study period.

In case of device-related adverse events or deficiencies (e.g., malfunction, usability issues, or inadequate labelling), the device will be returned to the sponsor for analysis, as per section 8.8.

9.2.1.2 Laboratory parameters

Not applicable.

9.2.1.3 Vital signs

Not applicable.

Assessments in subjects who prematurely stop the clinical investigation

Participants who withdraw from the investigation prematurely will undergo complete assessment of all scheduled evaluations up to the point of withdrawal, ensuring that valuable data is not lost while respecting participant autonomy. This approach maximizes the scientific value of the investigation while maintaining appropriate ethical standards for participant withdrawal. Safety follow-up will be conducted for 30 days post-withdrawal to monitor for any delayed adverse events or safety concerns that may be related to device use.

Follow-up of the subjects after the regular termination of the clinical investigation

Study completion involves comprehensive final assessment at 6 months, including all scheduled evaluations, device return, and final safety assessment. Participants will have the option to continue using the EpiSave device post-study, providing valuable long-term data on device performance and user satisfaction while ensuring continued access to seizure detection technology for participants who have benefited from the system.

9.3 Procedures at each visit

Screening Visit (Week -2 to Week 0)

The screening visit serves as the initial evaluation to determine participant eligibility and establish baseline characteristics for the investigation. This visit encompasses comprehensive medical history review to confirm epilepsy diagnosis, seizure history, and medication trials, ensuring that participants meet the

inclusion criteria for drug-resistant epilepsy and adequate seizure frequency. Eligibility assessment involves detailed evaluation of all inclusion and exclusion criteria, including device compatibility, caregiver availability, and technical requirements.

The informed consent process is conducted during this visit, providing participants and their families with comprehensive information about the investigation, including study procedures, potential risks and benefits, and participant rights. This process ensures that participants have complete understanding of the investigation before making the decision to participate. Baseline data collection includes documentation of current seizure frequency, medication regimen, quality of life assessment, and demographic characteristics that will serve as reference points for evaluating the EpiSave system's impact.

Device Setup Visit (Week 1)

The device setup visit focuses on providing the participant with the study materials, i.e., the connected watch with EpiSave and the camera for recordings. This visit aims to ensure that the participant fully understands how both devices work, so that the various procedures of the study are carried out correctly. Smartwatch and smartphone setup involves pairing the TicWatch Pro 5 with the participant's smartphone, installing the EpiSave application, and configuring the system with participant-specific settings and caregiver contact information. This setup process ensures that the system is properly configured for the participant's specific needs and environment.

Patient and caregiver training is comprehensive and includes instruction on device operation, alert system functionality, response procedures, and troubleshooting. This training ensures that both participants and caregivers understand how to use the EpiSave system effectively and respond appropriately to seizure alerts. System validation and testing involves verification that all system components are functioning properly, including accelerometry sensors, alert system, data transmission, and caregiver notification capabilities.

Confirmation of alert system functionality includes testing the dual-alert system to ensure that both participants and caregivers receive notifications appropriately. This testing verifies that the system can deliver alerts via SMS and push notifications, includes location information, and functions reliably in the participant's home environment.

Monthly Follow-up Visits (Weeks 4, 8, 12, 16, 20)

Monthly follow-up visits are scheduled by telephone and last approximately 5-10 minutes. The visits will be scheduled every 4 weeks, with a tolerance of approximately 5 days before or after, depending on patient availability.

These visits are conducted by a member of the clinical research team. Their primary purpose is to ensure ongoing assessment of participant compliance, device performance, and safety throughout the investigation period. Compliance assessment focuses on verifying device wear time, data transmission, and adherence to study procedures. This ensures that participants use the EpiSave system as intended and that the study meets its scientific objectives. The assessment includes reviewing device usage logs, battery charging patterns, and system uptime data.

Adverse event review involves systematic evaluation of any adverse events, serious adverse events, or device deficiencies that may have occurred since the last visit. This review ensures that any safety concerns are promptly identified, documented, and addressed according to regulatory requirements. Device performance evaluation assesses the EpiSave system's technical performance, including detection accuracy, alert functionality, and system reliability.

Data collection verification ensures that the EpiSave system is collecting and transmitting data appropriately, including seizure detection data, device usage information, and system performance metrics. This verification process ensures that the investigation maintains high data quality and that all necessary information is being captured for analysis of the device's effectiveness and safety profile.

10. SAFETY

10.1 Definition and Assessment of (Serious) Adverse Events and other safety related events

- **Adverse Event (AE)** (Art. 2 Abs 57 MDR)

Any untoward medical occurrence, unintended disease or injury or any untoward clinical signs (including an abnormal laboratory finding) in subjects, users or other persons whether or not related to the MD.

This comprehensive definition ensures that all potential safety concerns are captured and evaluated, regardless of their relationship to the EpiSave system or study procedures.

Adverse events are classified according to their causal relationship to the investigational device and study procedures. Device-related events are those causally related to EpiSave system use, including any adverse effects resulting from device malfunction, improper use, or system failure. Procedure-related events encompass those related to study procedures, such as device setup, training, or follow-up assessments. Underlying condition events are those related to epilepsy or other medical conditions that are not causally related to the device or study procedures. Unrelated events are those with no causal relationship to the device or procedures, representing coincidental medical events that occur during the study period.

- **Serious Adverse Event (SAE)** (Art. 2 Abs 58 MDR)

Any adverse event that led to any of the following:

(a) death,

(b) serious deterioration in the health of the subject that resulted in any of the following:

(i) life-threatening illness or injury,

(ii) permanent impairment of a body structure or a body function,

(iii) hospitalisation or prolongation of patient hospitalisation,

(iv) medical or surgical intervention to prevent life-threatening illness or injury or permanent impairment to a body structure or a body function,

(v) chronic disease,

(c) foetal distress, foetal death or a congenital physical or mental impairment or birth defect. Note: planned hospitalization for pre-existing condition, or a procedure required by the CIP, without a serious deterioration of the health status of the subject, is not considered an SAE

- **Device deficiency** (Art. 2 Abs 59 MDR)

Inadequacy of a medical device related to its identity, quality, durability, reliability, safety or performance, of an investigational device, including malfunction, user errors and inadequate information supplied by the manufacturer.

This definition encompasses both technical malfunctions of the EpiSave system and issues related to device use or performance that may affect safety or effectiveness.

- **Malfunction** (ISO14155)

Failure of an investigational device to perform in accordance with its intended purpose when used in accordance with the instructions for use or the CIP.

- **Device deficiency with Serious Adverse Event (SAE) potential** (Art. 80 Abs 1 letter c MDR; ISO14155)

Any device deficiency that might have led to a serious adverse event if appropriate action had not been taken, intervention had not occurred, or circumstances had been less fortunate.

- **Adverse Device Effect (ADE)** (ISO14155)

Adverse event possibly, probably or causally related to the use of an investigational device or procedures.

- **Serious Adverse Device Effect (SADE)** (ISO14155)

Adverse device effect (ADE) that has resulted in any of the consequences characteristic of a serious adverse event.

- **Unanticipated Serious Adverse Device Effect (USADE)** (ISO14155)

Serious adverse device effect (SADE) which by its nature, incidence, severity or outcome has not been identified in the current version of the risk analysis report.

Causal Relationship of SAE (MDCG 2020-10/1)

A causal relationship towards the medical device or the procedure of the investigation should be rated by the PI and the Sponsor as follows:

- **Not related:** The relationship to the device or procedures can be excluded.
- **Possible:** The relationship with the use of the investigational device is weak but cannot be ruled out completely. Alternative causes are also possible. Cases where relatedness cannot be assessed or no information has been obtained should also be classified as possible.
- **Probable:** The relationship with the use of the investigational device seems relevant and/or the event cannot reasonably be explained by another cause.
- **Causal relationship:** The serious event is associated with the investigational device or with procedures beyond reasonable doubt.

10.2 Adverse events categorization

The adverse events are categorized by the PI-Sponsor using the following algorithm:

Does the AE meet the seriousness criteria?

- No, it is not serious
 - Is the relationship to the device or the procedure possible, probable or causal?
 - No: non-related AE
 - Yes: ADE
- Yes, it is serious: SAE
 - Is the relationship to the device or the procedure possible, probable or causal?
 - No: non-related SAE
 - Yes: SADE
- Is it anticipated (within expected type, severity and frequency of the complications)?
 - No: unanticipated SADE (USADE)
 - Yes: anticipated SADE (ASADE)

10.3 Documentation and reporting in Medical Device Category C clinical investigations

Important note concerning all following sections of this chapter: add, respectively adapt to other local requirements in case of international investigations.

Foreseeable adverse events and anticipated adverse device effects

Foreseeable adverse events for the EpiSave system include false sense of security, where caregivers may become overly reliant on the device and reduce their vigilance for seizure events. System failure and missed seizures represent a critical safety concern, as device malfunction could lead to delayed intervention and increased SUDEP risk. False alarms and psychological impact may cause caregiver fatigue and reduce response effectiveness to genuine seizure alerts. Device biocompatibility issues in-

clude potential skin reactions, contact dermatitis, or allergic responses to smartwatch materials. Technical malfunctions encompass software failures, connectivity issues, or hardware problems that may affect device function. Caregiver response delays may occur when caregivers are unable to respond promptly to seizure alerts.

Anticipated device effects include skin irritation from device wear, which may occur due to prolonged contact with smartwatch materials. Battery life limitations may affect device performance and require frequent charging. Connectivity issues may impact data transmission and alert delivery. Alert fatigue may develop with frequent false alarms, potentially reducing caregiver responsiveness to genuine seizure alerts.

Documentation of device deficiencies and adverse events by the investigator

All device deficiencies (DD), all serious adverse events (SAE) and all non-serious adverse events (AE) OR non-serious adverse events (AE) identified in this CIP as being critical to the evaluation of the results of the investigation are collected, fully investigated and documented in the source document and appropriate CRF during the entire investigation period, i.e. from patient's informed consent until the last CIP-specific procedure, including a safety follow-up period.

- Documentation of AEs (including SAEs) by the PI includes diagnosis or symptoms, start and stop dates of event, event treatment, event resolution, assessment of seriousness and causal relationship to MD and/or investigation procedure (ISO14155:2020).
- Documentation of DDs by the PI includes description of event, start date, investigational device information, action taken with regard to the investigational device, and whether the DD led to an AE (ISO14155:2020).

Documentation requirements for adverse events and device deficiencies include complete AE/SAE report forms that capture all relevant information about the event, including onset, duration, severity, causality, and outcome. Device deficiency reports must document the nature of the deficiency, its impact on device function, and any corrective actions taken. Follow-up documentation ensures that events are monitored until resolution, including any ongoing effects or complications. Resolution documentation confirms the final outcome of the event and any long-term implications for participant safety or device performance.

Reporting of (Serious) Adverse Events, device deficiencies, and other safety related events

The reporting timeline for serious adverse events is within 24 hours to the sponsor and ethics committee, ensuring prompt notification and appropriate oversight. Device deficiencies are reported within 7 days to allow for thorough investigation and documentation. Adverse events are reported monthly through routine safety reporting procedures. This tiered reporting approach ensures that events with greater safety implications receive more urgent attention while maintaining comprehensive documentation of all safety events.

Reporting recipients include the sponsor, who is responsible for overall safety oversight and regulatory reporting, the ethics committee, which provides independent safety review and oversight, regulatory authorities, who require notification of serious events according to applicable regulations, and the Data Safety Monitoring Committee (DSMC), which provides independent safety monitoring and recommendations for study continuation or modification.

Reporting to the Sponsor:

The following events are to be reported to the Sponsor by the PI (or authorized designee) within 24 hours and 3 days (give reporting deadlines as applicable. These depend of stage of development and severity of possible consequences. Refer to the European guidance document MDCG 2020-10/1 for details) after becoming aware of the event:

All SAEs

Health hazards that require measures

DDs with SAE potential

Other AEs and DDs identified in this CIP as being critical to the evaluation of the results of the investigation

Other AEs and DDs identified in this CIP as being critical to the evaluation of the results of the investigation

The Sponsor shall define the timelines for reporting of non-serious AEs and DDs to the Sponsor.

The sponsor shall review the investigator's assessment of adverse events and determine and document in writing the seriousness and relationship to the investigational device and procedures required by the CIP (ISO 14155). The Sponsor shall evaluate AEs and SAEs with regard to causality and seriousness. Device deficiencies are also assessed regarding their potential to lead to an SAE (DD with SAE potential) (Art 32. ClinO-MD, ISO 14155:2020).

Pregnancies ...

Reporting to the Competent Ethics Committee and to Swissmedic:

The following events are to be reported to the CEC and to the CA promptly (Art. 33 ClinO-MD):

- a. any serious adverse event that has a causal relationship with the MD to be investigated, the comparator or the investigation procedure, or where such causal relationship is reasonably possible;
- b. any device deficiency that might have led to a SAE if appropriate action had not been taken, intervention had not occurred, or circumstances had been less fortunate; (DD with SAE potential);
- c. any new findings in relation to any event referred to in points (a) and (b).

In order to ensure prompt notification, the Sponsor may initially submit an incomplete notification. In line with European guidance document MDCG 2020-10/1 Rev. 1, reportable SAE and DD must be sent to Swissmedic within 7 days, or 2 days for SAE requiring prompt action for the safety of other study subjects.

If applicable: For conformity-related clinical trials in sub-categories C1 and C2 that are also being conducted abroad: The sponsor notifies the CEC and CA without delay of all events, device deficiencies and findings as specified above which arise from the conduct of the clinical trial abroad.

If safety and health hazards that require measures must be taken immediately during the conduct of the investigation, the Sponsor notifies the CEC and CA within 2 days of these measures and the circumstances which made them necessary (Art. 34 ClinO-MD).

If applicable: For clinical trials that are also being conducted or are also due to be conducted in EU or EEA states, The Sponsor notifies the CEC and CA within 2 days of all imposed or voluntary safety and protective measures that are being implemented in EU or EEA states and the circumstances that necessitated them (Art. 34 ClinO-MD).

Periodic safety reporting (Art. 35 ClinO-MD):

Once a year, the sponsor submits to the CEC and CA a list of all SAEs and DDs and provides it with a report on their severity, causal relationship with the device and/or the intervention and on the safety of the participants (outcome, event status). The sponsor informs the CEC and CA annually about the general progress and status of recruitment of the clinical investigation (also abroad). Any safety-relevant measures taken by the sponsor or imposed by ethics committees or authorities anywhere in the world as well as results from other clinical investigations with the investigational device (if applicable) shall be described. Based on the data presented in the report, the sponsor will draw his/her conclusions regarding the safety of the subjects and the continuation of the investigation. The safety report and the general progress report can be merged in one single report.

The cumulative list of reportable serious adverse events and device deficiencies (MDCG 2020-10/2 Rev. 1) per cut-off date is submitted in parallel.

Follow-up of (Serious) Adverse Events

Follow-up requirements for serious adverse events include complete resolution documentation that confirms the final outcome of the event and any ongoing effects. Ongoing monitoring continues until resolution to ensure that any delayed effects or complications are identified and managed appropriately. Final outcome assessment provides comprehensive documentation of the event's impact on participant safety and device performance, enabling appropriate regulatory reporting and study oversight.

10.4 Documentation and reporting in Medical Device Category C clinical investigations

If the permitted dose guidance value is exceeded at any time, the Sponsor or the PI notifies the CEC within 7 working days after becoming aware of the event [Art. 39 ClinO-MD].

f in the case of Category C clinical investigations with MDs that emit ionising radiation, if the permitted dose guidance value is exceeded at any time, the Sponsor or the PI notifies the CA within 7 working

days after becoming aware of the event [Art. 39 ClinO-MD].

Within one year of the completion or discontinuation of the investigation involving examinations with radioactive sources, the Sponsor sends to the CEC and to the CA a final report containing all information relevant to radiation protection, in particular an estimate of the doses to which the subjects were exposed [Art. 39 ClinO-MD].

11. STATISTICAL METHODS

11.1 Hypothesis

The primary hypothesis states that EpiSave will demonstrate $\geq 90\%$ sensitivity for video-confirmed nocturnal generalized convulsive seizure (GCS) detection in real-world ambulatory settings. This hypothesis is based on the Phase 2 validation results demonstrating 96% sensitivity in controlled epilepsy monitoring unit settings, with the expectation that real-world performance will be slightly lower due to environmental factors and device wear compliance variations.

Secondary hypotheses encompass multiple dimensions of device performance and clinical impact. EpiSave will maintain a false alarm rate of less than 1 per week in ambulatory settings, ensuring that the device provides reliable seizure detection without excessive false alarms that could lead to caregiver fatigue. Caregiver response times will be ≤ 5 minutes for 90% of seizure alerts, ensuring timely intervention for detected seizure events. System uptime will be $\geq 95\%$ over the study period, demonstrating reliable device operation and data collection. Patient and caregiver satisfaction scores will be $\geq 4.0/5.0$ on standardized scales, indicating high acceptability and usability of the EpiSave system.

11.2 Determination of Sample Size

The target sample size is a minimum of 75 verified nocturnal GCS events in at least 50 people with epilepsy (PWE). We plan to recruit between 77 to 100 patients in order to reach the 75 GCS in 50 people. These figures correspond to the standards proposed for phase 4 studies of seizure detection devices in ambulatory patients. We anticipate that 65% to 80% of recruited patients will eventually suffer a nocturnal GCS for which both video recordings and EpiSave data will be available for analysis, translating into the recruitment of 77 to 100 PWE over 30 months (i.e. ≤ 1 PWE/4 months/center).

This sample size determination is based on literature recommendations for phase 4 seizure detection studies and provides 90% statistical power to detect sensitivity $\geq 90\%$ with 95% confidence. The event-driven design ensures adequate statistical power regardless of recruitment timeline variations, as the analysis is based on the number of seizure events rather than the number of participants.

The statistical rationale incorporates literature-based recommendations from published phase 4 seizure detection studies, power calculations that provide 90% power to detect sensitivity $\geq 90\%$ with 95% confidence, an event-driven design that ensures adequate statistical power regardless of recruitment timeline, and dropout consideration that accounts for a 20% anticipated dropout rate. The recruitment timeline is set at 30 months with a target of ≤ 1 PWE per 4 months per center, ensuring realistic recruitment expectations while maintaining study feasibility.

11.3 Statistical criteria of termination of the investigation

Early termination criteria are established to ensure participant safety and study integrity while maintaining scientific rigor. Safety concerns identified by the Data Safety Monitoring Committee (DSMC) may warrant early termination if the risk-benefit ratio becomes unfavorable. Regulatory authority decisions may require study termination based on safety or regulatory concerns. Insufficient recruitment after 30 months may necessitate early termination if the study cannot achieve its scientific objectives within a reasonable timeframe. Study objectives may be achieved early if the primary endpoint demonstrates clear superiority or if safety concerns arise that outweigh potential benefits.

11.4 Planned Analyses

Datasets to be analysed, analysis populations

Three analysis populations will be defined to ensure comprehensive evaluation of the EpiSave system's performance. The Per-Protocol (PP) population includes participants completing the study with protocol compliance, providing the most rigorous evaluation of device effectiveness under optimal conditions. The Intention-to-Treat (ITT) population includes all enrolled participants regardless of completion status, ensuring that the analysis reflects real-world usage patterns and includes all participants who received the device. The Safety Population includes all participants receiving the device, providing comprehensive safety evaluation regardless of protocol compliance or study completion.

Primary Analysis

The primary endpoint analysis focuses on EpiSave sensitivity for video-confirmed nocturnal GCS detection. Sensitivity calculation will determine the proportion of true nocturnal GCS detected by EpiSave out of all video-confirmed nocturnal GCS events. Confidence intervals will be calculated using 95% Wilson confidence intervals to provide appropriate statistical precision. The statistical test will employ a one-sample proportion test against the 90% target sensitivity, with the analysis population being the per-protocol population including all participants with at least one nocturnal GCS event.

Secondary Analyses

Secondary analyses encompass multiple dimensions of device performance and clinical impact. False alarm rate analysis will evaluate false alarms per patient per day using negative binomial regression for count data, with random effects for age, seizure frequency, and device wear time, targeting less than 1 false alarm per week per patient. Caregiver response time analysis will assess time from alert to caregiver response using Kaplan-Meier survival analysis, right-censored at 30 minutes, targeting ≤ 5 minutes for 90% of alerts. System reliability analysis will evaluate system uptime percentage using mixed-effects models for repeated measures with participant and time as random effects, targeting $\geq 95\%$ uptime over study duration.

Interim analyses

The interim analysis plan includes timing after 25%, 50%, and 75% of target events to provide ongoing safety and efficacy monitoring. The primary focus is safety and futility assessment to ensure participant safety and study integrity. The statistical method employs O'Brien-Fleming stopping boundaries to maintain appropriate statistical rigor while allowing for early termination if safety concerns arise or if the study objectives are clearly achieved or unachievable. The DSMC will conduct independent review of interim results to ensure objective evaluation and appropriate recommendations for study continuation or modification.

Deviation(s) from the original statistical plan

Any deviations from the original statistical plan will be documented and justified to ensure transparency and maintain scientific integrity. Protocol deviations will be assessed for their impact on study validity and regulatory compliance, with appropriate amendments submitted to regulatory authorities and ethics committees as required.

11.5 Handling of missing data and drop-outs

The primary analysis will employ complete case analysis to ensure that results are based on participants with complete data. Sensitivity analysis will utilize multiple imputation for missing data to assess the robustness of results to missing data assumptions. Drop-out analysis will compare completers versus drop-outs to identify any systematic differences that may affect study validity.

Drop-out management includes prevention through regular follow-up and support to minimize participant withdrawal, documentation of reasons for drop-out and data completeness to understand the impact of withdrawals, and intent-to-treat analysis including all randomized participants to ensure that results reflect real-world usage patterns and maintain study validity.

12. QUALITY ASSURANCE AND CONTROL

Quality assurance and control systems are implemented to ensure the integrity, reliability, and regulatory compliance of the clinical investigation. The Sponsor is responsible for implementing and maintaining comprehensive quality assurance and quality control systems with written Standard Operating Procedures (SOPs) and Working Instructions at all investigational sites. The Principal Investigator (PI) is responsible for proper training of all involved investigation personnel to ensure adherence to protocol requirements and regulatory standards.

12.1 Data handling and record keeping / archiving

Data handling and record keeping procedures ensure that all investigation-related documents are properly maintained and archived according to regulatory requirements. A comprehensive list of essential clinical investigation documents that must be maintained at the investigation site and sponsor file is provided in ISO 14155 Annex E, ensuring complete documentation of all study activities and regulatory compliance.

Case Report Forms

Investigation data is recorded using electronic Case Report Forms (eCRFs) that provide comprehensive documentation of all study variables and endpoints. A CRF is maintained for each enrolled subject and must be kept current to reflect subject status at each phase during the course of the investigation. Subjects are not identified in the CRF by name, initials, or birth date, ensuring confidentiality and privacy protection.

The coding system used for the investigation employs subject numbers in combination with year of birth, following the guidance document published on [swissethics.ch](https://swissethics.ch/assets/Themen/akzeptierte_verschluesselung_e.pdf) regarding "coding of trial subject accepted by swissethics and secure storage of subject identification list" (https://swissethics.ch/assets/Themen/akzeptierte_verschluesselung_e.pdf). This coding system ensures participant confidentiality while maintaining the ability to link data to specific participants when necessary for safety monitoring or regulatory requirements.

The eCRF design incorporates built-in validation rules and range checks to ensure data quality and completeness. Real-time data entry with immediate validation provides instant feedback on data quality and enables prompt correction of any data entry errors. The person(s) authorized by the PI to enter data in the CRF are listed on the delegation log, ensuring accountability and proper authorization for data entry activities.

Specification of source data and source documents

Source data must be available at the site to document the existence of the investigation subjects and include original documents relating to the investigation, as well as medical treatment and medical history of the subject. For electronic source data from the EpiSave application and automatic recording devices, data is handled, transferred, stored, and accessed by the PI and authorized staff through secure, encrypted systems that maintain data integrity and confidentiality.

Source documents in this investigation include medical records documenting epilepsy diagnosis, seizure history, and medication trials; device data logs from the EpiSave system capturing seizure detection data, device usage, and system performance; video recordings from autonomous camera systems providing ground truth for seizure confirmation; patient diaries documenting seizure events and device usage; and laboratory reports for any relevant examinations. Source data verification involves 100% verification of critical data elements to ensure accuracy and completeness of the database.

Original source data are kept at the investigational site in secure, access-controlled locations, with electronic data stored in encrypted databases with appropriate backup systems. The source data description and source data location are detailed in a separate document in the Appendices to provide comprehensive documentation of all data sources and their locations.

Archiving of essential clinical investigation documents

All documents of the investigation must be archived for a minimum of 10 years after regular or premature termination of the investigation, with 15 years required for implantable devices in Switzerland (Art. 40 ClinO-MD). The Sponsor (Art. 40 Abs 1 ClinO-MD) is responsible for maintaining comprehensive archives of all study documents, while the PI (Art. 40 Abs 2 ClinO-MD) is responsible for maintaining site-specific documentation and ensuring proper archiving procedures.

Archiving procedures specify the location and length of storage, with electronic documents stored in secure, encrypted databases with appropriate backup systems, and physical documents stored in secure, access-controlled facilities. The archiving system ensures long-term preservation of all essential clinical investigation documents while maintaining confidentiality and regulatory compliance.

12.2 Data management

Data management procedures encompass comprehensive plans for data entry, coding, security, and storage, including processes to promote data quality such as range checks for data values and validation procedures. For electronic data capture systems, this includes detailed procedures for verification, validation, and securing the database to ensure data integrity and regulatory compliance.

Data will be stored in coded form after statistical analysis, maintaining participant confidentiality while preserving the ability to link data to specific participants when necessary for safety monitoring or regulatory requirements. If data is anonymized, the anonymization process will be documented to ensure that it cannot be reversed and that participant privacy is protected. Detailed data management procedures are documented in separate Standard Operating Procedures referenced in the CIP.

Data Management System

The investigation utilizes an Electronic Data Capture (EDC) system that provides real-time data entry, validation, and monitoring capabilities. The system features encrypted data transmission and storage, daily automated backups, and comprehensive security measures to ensure data integrity and confidentiality. The system is hosted in secure, compliant data centers with appropriate redundancy and disaster recovery capabilities.

The EDC system is tested before the investigation begins to ensure proper functionality, security, and compliance with regulatory requirements. System testing includes validation of data entry procedures, security measures, backup systems, and user access controls to ensure reliable operation throughout the investigation period.

Data security, access and back-up

Data security measures include user authentication and authorization systems, encrypted data transmission using AES-256 encryption, secure data storage in compliant data centers, and regular security audits to ensure ongoing protection. Access control is implemented through role-based access permissions that limit data access to authorized personnel based on their responsibilities and training.

An audit trail is maintained for all data access, providing comprehensive documentation of who accessed data, when, and for what purpose. Regular access reviews are conducted to ensure that access permissions remain appropriate and that unauthorized access is prevented. Backup systems include daily automated backups with off-site storage to ensure data recovery capabilities in case of system failure or disaster.

Analysis and archiving

Data extraction procedures ensure that data are properly extracted from the EDC system for statistical analysis using validated software. Database status recording provides comprehensive documentation of all data processing activities, including extraction dates, analysis procedures, and data transformations. The duration and place of storage are specified to ensure long-term preservation and regulatory compliance.

Statistical analysis is conducted using validated software to ensure accuracy and reproducibility of results. Long-term archiving includes storage of all study data and documents for the required retention period of 15 years as per regulatory requirements, ensuring that all data and documentation are preserved for potential future analysis or regulatory review.

Electronic and central data validation

Data validation procedures include range checks for all numeric variables to ensure that values fall within expected ranges, consistency checks across related variables to identify potential data entry errors, completeness checks for required fields to ensure that all necessary data are captured, and cross-validation with source documents to verify data accuracy.

These validation procedures ensure that the database maintains high quality and integrity throughout the investigation, enabling reliable statistical analysis and regulatory compliance. Any data quality issues

identified through validation procedures are documented and resolved according to established procedures.

12.3 Monitoring

Regular monitoring visits are conducted at the PI's site prior to the start and during the course of the investigation, organized by the Sponsor. Monitoring activities include comprehensive review of subject enrollment logs, informed consents and informed consent process, source data verification, inclusion and exclusion criteria compliance, subjects' visit schedule adherence, safety monitoring, processing of subjects' data, preservation of subjects' confidentiality, reporting to CEC and regulatory authorities, and provisions of records and data retention.

The extent and nature of monitoring activities are detailed in a comprehensive Monitoring Plan that is annexed to the CIP (Annex XV, Chapter 2, Art. 3.6.6. MDR). The monitoring is conducted by qualified clinical research associates with appropriate training and experience in medical device investigations. Monitor qualification and training requirements include certification in Good Clinical Practice (GCP), medical device regulations, and specific training on the EpiSave system and study procedures.

Procedures for reviewing monitoring visit reports include comprehensive documentation of all findings, follow-up on monitoring findings with appropriate corrective actions, and regular communication between monitors, investigators, and the sponsor to ensure ongoing compliance and quality improvement. Source data and documents are accessible to monitors, and questions are answered during monitoring by the PI and site staff to ensure complete understanding and compliance with study requirements.

12.4 Audits and Inspections

Auditing procedures include regular internal audits conducted by the Sponsor to ensure compliance with protocol requirements and regulatory standards. The audit process is independent from the PI and Sponsor to ensure objective evaluation of study conduct and data integrity. External audits may be conducted by regulatory authorities or independent auditors as required.

The frequency and procedures for auditing are specified in the Quality Management Plan, with regular audits conducted throughout the investigation period. Documentation of the investigation and source data/documents are accessible to auditors and inspectors, including CEC and regulatory authorities, and questions are answered during inspections by appropriate study personnel. All involved parties maintain strict confidentiality of subject data throughout the audit and inspection process.

12.5 Confidentiality, Data Protection

Data protection measures ensure that participant confidentiality is maintained throughout the investigation while allowing appropriate access for monitoring, audits, and inspections. Direct access to source documents is permitted for purposes of monitoring (chapter 12.3), audits and inspections (chapter 12.4) to ensure regulatory compliance and data integrity.

Access to investigation documents, datasets, and randomization codes is restricted to authorized personnel with appropriate training and confidentiality agreements. During the investigation, access is limited to study personnel, monitors, auditors, and regulatory authorities as required. After the investigation, data access is maintained according to regulatory requirements and participant consent for long-term data storage and potential future research use.

12.6 Storage of biological material and related health data

This investigation does not involve collection or storage of biological material. However, coded data from subjects who have consented for further use of their data (independently of the investigation-specific consent) may be stored in a registry for an undetermined length of time and could be re-used for other research projects, provided previous approval by the CEC is obtained.

Personal data storage duration and location are specified according to regulatory requirements, with data anonymized after the end of the storage period to ensure participant privacy protection. The information provided here matches the information given in chapters 8.4 and 9.2.5 to ensure consistency

throughout the protocol. Coded data are only stored if subjects' consent for further use has been obtained, with this consent given independently of participation in the investigation (Art. 17. ClinO).

13. PUBLICATION AND DISSEMINATION POLICY

The publication policy for the results of this clinical investigation is established according to Art. 42 ClinO-MD to ensure transparent and timely dissemination of study findings. The Sponsor is committed to communicating the results of the investigation to subjects, healthcare professionals, the public, and other relevant groups through multiple channels to maximize the clinical and scientific impact of the study.

13.1 Communication Strategy

Results will be communicated through a comprehensive dissemination strategy that includes publication of peer-reviewed manuscripts in high-impact medical journals, presentation at international epilepsy and medical device conferences, development of a lay language summary for participants and the general public, and reporting in recognized clinical trial results databases. The communication strategy ensures that findings reach diverse audiences including clinicians, researchers, patients, caregivers, and regulatory authorities.

Authorship eligibility guidelines follow the International Committee of Medical Journal Editors (ICMJE) criteria, ensuring that all individuals who have made substantial contributions to the conception, design, execution, or interpretation of the study are appropriately recognized. Professional medical writers may be engaged to assist with manuscript preparation, but all content will be reviewed and approved by the study investigators and Sponsor. The Sponsor retains ultimate authority over publication decisions, data sharing arrangements, and dissemination activities.

13.2 Data Sharing and Access

Plans for granting public access to the full Clinical Investigation Plan, subject-level dataset, and statistical code are established to promote scientific transparency and enable secondary research. Subject-level data will be de-identified and made available through appropriate data sharing platforms, with access granted to qualified researchers who submit scientifically sound research proposals. Statistical code will be made publicly available to enable replication of analyses and promote methodological transparency.

Protection of trade secrets is maintained through appropriate data sharing agreements that protect proprietary algorithms and device specifications while enabling scientific validation and replication. The Sponsor will work with regulatory authorities and ethics committees to ensure that data sharing arrangements comply with applicable regulations and participant consent requirements.

13.3 Regulatory Reporting Requirements

The Sponsor will enter and publish a summary of the results of the clinical investigation in a publicly recognized register as specified in Art. 64 Abs. 1 lit a or b ClinO. For this investigation involving an investigational medical device, the summary will be published at the latest before the device is placed on the market or one year after submitting the final report if the device has not been placed on the market by this point in time, in accordance with Article 37.

The Sponsor also ensures that a lay summary of the results is entered in BASEC within the period specified above. The entry is made at least in the national languages of Switzerland in which the study participants were recruited, ensuring accessibility to all participants and the Swiss public.

13.4 Participant Communication

The investigator will provide each participant with the lay summary of the results of the clinical investigation at the end of the study, directly and in a format that is easily understood. The investigator ensures

that participants are adequately informed about this in the patient information document and are informed where the lay summary of the results of the clinical investigation will be published online.

If publication of the results is not possible within the specified period for scientific reasons, the Sponsor will explain this in the application documents and indicate when publication will take place, ensuring transparency about any delays in result dissemination.

14. FUNDING AND SUPPORT

14.1 Funding

Provide brief statement of sources and types of financial support for the investigation. If applicable, reference to other places or contracts/documents where this information is captured.

14.2 Other Support

Provide brief statement of any other type of support received to conduct the investigation (MD, comparator, investigation material, software's, ...). If applicable, reference to other places or contracts/documents where this information is captured.

15. INSURANCE

Give proof of insurance cover or indemnification of subjects in case of injury, pursuant to Art. 3 ClinO-MD. E.g., "Insurance is provided by the Sponsor and fulfils the legal provision of art. 3 ClinO-MD. A copy of the insurance certificate is filed in Investigator's file and in the Sponsor's file."

Note: Category A1 performance studies are exempt from liability coverage requirements (ClinO Art. 12). Categories A2 and C performance studies need to document the guarantee of liability (insurance certificate or equivalent guarantee) (ClinO Art. 13).

The insurance must cover damage occurring up to 20 years after the end of the clinical investigation.

The policy value shall be set in accordance with ClinO Annex 2.

It can be referred here to another place where the document is found, e.g., chapter 17 or elsewhere.

16. REFERENCES

Provide a list of the references pertaining and cited in the CIP.

1. Declaration of Helsinki, Version October 2013 (<http://www.wma.net>)
2. Humanforschungsgesetz, HFG Bundesgesetz über die Forschung am Menschen (Bundesgesetz über die Forschung am Menschen, HFG) vom 30. September 2011/ Loi fédérale relative à la recherche sur l'être humain (loi relative à la recherche sur l'être humain, LRH) du 30 septembre 2011 / Legge federale concernente la ricerca sull'essere umano (Legge sulla ricerca umana, LRUm) del 30 settembre 2011
3. Verordnung über klinische Versuche mit Medizinprodukten (KlinV-Mep) vom 1. Juli 2020 / Ordonnance sur les essais cliniques de dispositifs médicaux (OClin-Dim) du 1er juillet 2020 /. Ordinanza sulle sperimentazioni cliniche con dispositivi medici (OSRUM-Dmed) del 1 luglio 2020
4. Verordnung über klinische Versuche mit Ausnahme klinischer Versuche mit Medizinprodukten (Verordnung über klinische Versuche, KlinV) vom 20. September 2013 / Ordonnance sur les essais cliniques hors essais cliniques de dispositifs médicaux (Ordonnance sur les essais cliniques, OClin) du 20 septembre 2013. Ordinanza sulle sperimentazioni cliniche ad eccezione delle sperimentazioni cliniche con dispositivi medici (Ordinanza sulle sperimentazioni cliniche, OSRUM) del 20 settembre 2013
5. Medizinprodukteverordnung (MepV) vom 17. Oktober 2001 / Ordonnance sur les dispositifs médicaux (ODim) du 17 octobre 2001 / Ordinanza relativa ai dispositivi medici (ODmed) del 17 ottobre 2001
6. Medical Device Regulation (EU) 2017/745 of 5 April 2017 (MDR)
7. MDCG 2024-3 Guidance on content of the Clinical Investigation Plan for clinical investigations of medical devices
8. MDCG 2020-10/1 Safety reporting in clinical investigations of medical devices under the Regulation (EU) 2017/745 (https://ec.europa.eu/health/sites/health/files/md_sec-tor/docs/md_mdcg_2020-10-1_guidance_safety_reporting_en.pdf)
9. EN ISO 14155: Clinical investigation of medical devices for human subjects - Good clinical practice (www.iso.org)
10. EN ISO 10993: Biological evaluation of medical devices (www.iso.org)
11. EN ISO 14971: Application of risk management to medical devices (www.iso.org)
12. WHO, International Clinical Trials Registry Platform (ICTRP) (<http://www.who.int/ictrp/en/>)
13. Strahlenschutzverordnung (StSV) vom 26. April 2017 / Ordonnance sur la radioprotection (ORaP) du 26 avril 2017 / Ordinanza sulla radioprotezione (ORaP) del 26 aprile 2017.
14. International Conference on Harmonization (ICH) Guideline for Good Clinical Practice E6(R2), (www.ich.org).
15. Zz
16. Yy

17. APPENDICES

NOTE: Further relevant information can be found in the ISO14155, Annex A Clinical Investigation Plan (CIP)

Documents that do frequently change during the course of the investigation can be mentioned as 'documents provided separately' and listed here.

The section headings can be renamed accordingly.

Investigator's Brochure

General Insurance Conditions, insurance certificate

List of norms

List of investigational sites / PIs (List of countries or centres where data will be collected)

Specific protocols (e.g. MRI)

Case Report Form (ISO14155 Annex C)

Monitoring Plan

Other material handed over to the patients